

2. The Bases for Freedom in Society

Can two walk together, except they be agreed?

—Amos 3:3

During summer months, a roadside stand outside Blacksburg displays seasonal fruits and vegetables. I can purchase watermelons in quantities that I choose at prices which, by convention, are established by the salesman. There is little or no haggling, and a transaction can be completed in seconds. Economic exchanges like this are so familiar to us, so much a part of everyday routine, that we often overlook the bases upon which such institutions rest. I do not know the fruit salesman personally, and I have no particular interest in his well-being. He reciprocates this attitude. I do not know, and have no need to know, whether he is in direst poverty, extremely wealthy, or somewhere in between. Likewise, his ignorance concerning my economic status is complete. Yet the two of us are able to complete an exchange expeditiously, an exchange that both of us accept as “just.” I make no effort to seize watermelons without his consent and without payment. The vendor does not grab coins and currency from my purse.

We transact exchanges efficiently because both parties agree on the property rights relevant to them. Both of us acknowledge that the watermelons, stacked neatly by the roadside, are “owned” by the salesman, or by the person or firm for whom he acts as agent. Both of us also acknowledge that I have the rights of disposition over the money in my pockets or in my bank account. Furthermore, both of us recognize that any unilateral attempt to violate these assigned rights of exclusion will be subject to penalty through the arms and agencies of the state. In other words, both of us agree on what “the law” is that is relevant to the exchange in question.

Let us now consider a descriptively similar example, but one that is quite

different in essential respects. Suppose that it is commonly known that the watermelons are grown by a farmer who stacks them alongside the road available for the taking by passersby without charge. Suppose that, under these conditions, I happen upon someone who tries to exact a money price from me. This setting is quite different behaviorally from the first. Since I do not recognize any right of ownership on the part of the person who has now expropriated the melons, I am reluctant to pay, despite the fact that my evaluation may exceed the money price asked. On the other hand, the nominal possessor of the goods is unwilling to meet my demands without a price since, in his view, a right of ownership has been established. Ordinary exchange, which seemed so simple and straightforward in the first example, is made extremely difficult here because the parties do not agree on the "law of property" in being, which means, in turn, that they are uncertain as to what action the state might take in any dispute. If I should be certain that the alleged salesman has no rights under law, I should take the watermelons, and should he try to prevent my doing so, I should call the police. But, of course, in an acknowledged absence of legal rights, he would not exert physical force to constrain me. On the other hand, should I be certain that his alleged rights were enforceable, then no matter how I might feel about the morality of his position, I should probably purchase the melons and go on my way as before.

The point illustrated by these simple examples is clear. Economic exchange among persons is facilitated by mutual agreement on defined rights. Both parts of this principle must be satisfied. Individual rights must be well defined and nonarbitrary, and, in addition, these rights must be recognized and accepted by participants. If rights are known to be well defined and nonarbitrary but if knowledge about them is available to persons only on considerable investment in information gathering, many exchanges that are otherwise mutually beneficial may never come into being. Once both parts of the principle are met, however, once the limits of each person's rights are defined by agreement, economic interchange becomes almost the archetype of ordered anarchy. Individuals can deal with one another through wholly voluntary behavior without coercion or threat. They can enter into and complete exchanges without detailed knowledge of the political persuasions, sexual attitudes, or economic statuses of their actual trading partners. The traders may be unequal in any or all of such descriptive characteristics, yet they can and do deal with one another as equals in the exchange itself. In this classic

sense, economic exchange is wholly impersonal, which seems to be precisely the ideal-type interaction embodied in ordered anarchy. Each person is treated strictly as he is, and presumably as he wants to be, in such a relationship. The fruit stand operator may beat his horse, shoot dogs, and eat rats. But none of these qualities need affect my strictly economic trade with him.

Under regimes where individual rights to do things are well defined and recognized, the free market offers maximal scope for private, personal eccentricity, for individual freedom in its most elementary meaning. The failure of the romantic advocates of anarchy to recognize this feature of free markets is difficult to understand; this is one of the sources for the paradox observed in the 1970s and noted above in Chapter 1. Socialist organization, defined broadly as extended collective or state control over voluntary exchange processes, must be and can only be antithetic to anarchy, despite the surprising linkage of these two contradictory organizational norms in much of the romantic literature.¹

Economics, the science of markets or of exchange institutions, commences with a well-defined structure or set of individual rights and offers explanatory, predictive propositions concerning the characteristics of outcomes along with conditional predictions about the effects of imposed structural changes on such outcomes.² Economic theory is sufficiently powerful to explain many varieties of exchange relationships. The central tradition of this theory involves analysis of two-party exchanges imbedded in and constrained by a network of interrelated potential and actual trading pairs. It is not my pur-

1. For a relatively recent book that falls within the romantic tradition, and which summarizes other works, see Daniel Guérin, *Anarchism*, intro. Noam Chomsky, trans. Mary Klopfer (New York: Monthly Review Press, 1970).

There exists a variant of anarchism which is, instead, based squarely on the recognition of the free market's role in facilitating individual relationships in the absence of government. This has been called "private property anarchism" by Laurence Moss, who traces out the contributions of Americans to this variant. See Laurence S. Moss, "Private Property Anarchism: An American Variant" (Paper presented at the Southern Economic Association meeting in Washington, D.C., November 1972). Murray Rothbard is a modern expositor of this variant. See his *For a New Liberty*.

2. This is my own definition of economics. It is elaborated in some detail in my 1963 presidential address to the Southern Economic Association. See "What Should Economists Do?" *Southern Economic Journal* 30 (January 1964): 213–22. This view is at variance with those who define economics in terms of the central maximizing principle. Differences in definition here need not, however, affect the main argument of the text.

pose to summarize this theory here. As Chapter 3 will suggest, however, straightforward extension of some of the models to the many-party, complex exchanges that collective action may represent yields substantial explanatory results. In all cases, and this is my emphasis, the analysis proceeds from some initial imputation of rights among participating persons in the interchange, an imputation that is assumed to be conceptually observable and also to be recognized and respected by all parties.

Commonality and Noneconomic Interaction

The dependence of efficient trade upon a delineation and identification of individual rights is revealed most clearly in the case of fully partitionable “private” goods. It should be evident that the requirements for mutually accepted definitions of structure need not be so restricted. Consider facilities that may, due to either technological necessity or social decision, be accessible to all members of the relevant group. Mutual agreement on the behavioral limits with respect to the use of such property is, at base, no different from mutual agreement on the boundary lines for strictly private holdings. My recognition that the fruit salesman owns the watermelons is not, in concept, different from my recognition that both he and I have rights to walk along the village street, rights that both of us honor and respect, and which neither of us calls into dispute. We both utilize the common-access facility, and we do so without overt conflict only because of our mutual recognition and acceptance of these rights. If a particular road or street should be “Private”; if one of us should hold legal title to the facility which embodies rights to exclude, the other would normally respect a “No Trespassing” sign if one appears. This despite the possibility that, descriptively, the road or street facility might be identical in the two settings. Conflict emerges, or may emerge, not from any specific assignment of individual rights, but from disagreement over and uncertainty about just what the legally enforceable assignment is.

The same principle applies to many aspects of human behavior that are not normally classified as “economic” and that are not explicitly treated as exchange relationships. The reconciliation of individuals’ desires to “do their own things” with the fact that they live together in society is accomplished largely by mutual agreement on spheres of allowable or tolerated activity. “Equal freedom,” as a norm or rule for social intercourse, has little or no

meaning until and unless individuals are first identified in terms of acknowledged limits to behavior. The acceptance of such limits is so familiar to us all, pervading wide reaches of routine behavior as well as our attitudes toward the behavior of others, that we rarely think of the structure of individual "rights" that is underneath. Our attention is turned to definition of rights only when the tolerated limits are exceeded, when previously accepted boundaries are crossed. Only at this stage do we begin to consider drawing the limits more carefully, possibly calling on enforcement agents, or thinking about recourse to personal means of redress or defense.

The set of manners, the customary modes for personal behavior, which reflects the mutual acceptance of limits, will of course vary somewhat from culture to culture, but it is relatively easy to think of examples in any setting. I do not start my power mower early on Sunday morning, and my neighbor does not play stereo music loudly after eleven at night. Both of us recognize the possibly harmful effects on the other, and we refrain from imposing costs in this manner, even at some personal sacrifice. If one of us violates this set of "live and let live" rules, the other is prompted to take specific action in redress. If my neighbor operates his stereo loudly in the wee hours, and does this repeatedly, I should be prompted to try deliberately to annoy him with my lawn mower, to get the police to enforce the antinoise or antinuisance ordinance, or, if one does not exist, to try to get the town council to enact such an ordinance. If all else fails, I might then resort to direct physical action against the neighbor's property or person.

It is in this context that some of the behavioral changes of the 1960s raise fundamental and disturbing issues for social stability. As noted, individuals have lived, one with another, under implicit behavioral rules that were respected by all, or nearly all, persons in community. But one of the instruments employed by the participants in the counterculture involved the explicit flaunting of traditional codes of conduct, the direct and open disregard for what had previously been considered to be acceptable standards for elementary "good manners." This placed stresses on the ordered anarchy that still describes much of ordinary social life in our society, stresses which were evidenced by calls for "law and order," for formalization and enforcement of rules that were previously nonexistent.

Social stability requires agreement on and enforcement of a structure of individual rights, whether these rights be over the disposition of privately

partitionable goods, over the usage of common facilities, or over ordinary patterns of interpersonal behavior, and whether the enforcement be externally imposed or internally monitored. A possible source of ambiguity should be mentioned here. Mutual agreement, backed up if necessary by effective enforcement, is a necessary condition for social interchange. But this agreement may embody any one of an almost infinite variety of actual distributions and/or imputations of rights among persons, constrained by any one of an equally large number of possible sets of rules for personal behavior. Neither the specific distribution of rights among separate persons nor the general characteristic of the rights structure itself is relevant directly to the issue of mutual agreement, certainty in definition, and enforcement.³ Physical facilities may be variously partitioned among individuals, as units of "private property." Or, alternatively, such facilities may be organized under rules that dictate common usage by large groups of persons, including the whole membership of the community. If the limits to individual behavior are well defined, voluntary social interaction can proceed in an orderly fashion under any structure. Interpersonal dealings can take place under *any* agreed-on assignment. Only at a second, and quite different, level of discourse do problems arise about the relative desirability of specific distributions of rights and/or about the relative efficiencies of different structures. The tendency or proclivity to inject either equity or efficiency norms, or both, too early has plagued the discussion of property rights throughout the ages.

It is, of course, possible to evaluate alternative distributions on personally chosen criteria of "justice" or "equity." More positively, it may be possible to array alternative structures of rights by criteria of economic efficiency. The owners of cattle and sheep on a grazing range may either (1) hold rights to the common pasture or (2) hold individually fenced-off parcels of land. The second arrangement may, on examination and analysis, prove to be more efficient than the first in some standard efficiency sense, but the relative inef-

3. This is emphasized clearly by David Hume; see *A Treatise of Human Nature*, ed. L. A. Selby-Bigge (Oxford: Clarendon Press, 1960), pp. 502–3. Hume's whole discussion concerning the origins of property rights and the advantages of such rights for social stability is similar in many respects to that which is developed in this book.

Hegel's basic conception of property is also similar to that developed here. See Shlomo Avineri, *Hegel's Theory of the Modern State* (Cambridge: Cambridge University Press, 1972), pp. 88f.

iciency generated under the common-ownership arrangement is not at all comparable and indeed is different in kind from that which might emerge when mutuality of agreement on rights disappears, when there is uncertainty as to just what structure of rights will be legally enforced. The range wars in the American West in the late 1880s arose because of such uncertainties as these, and not because the structure in existence was grossly inefficient in the orthodox sense or was demonstrably “unjust” by other criteria.⁴

Rights and Contract

The argument to this point is straightforward and could scarcely be disputed. A necessary starting position for a society of free individuals, related one to another in a network of interdependence, is some agreement on a structure of rights which, in effect, defines the entities who enter negotiations. It is difficult even to imagine a relationship when such mutual agreement is wholly absent. How could two men, meeting for the first time, carry out the simplest form of interchange without implicit acceptance of some behavioral limits?

As I have noted, these bases for individual behavior in society are simply taken for granted by most of us, so much so that we rarely pause to examine the issues that are posed. As Blackstone suggested: “These inquiries, it must be owned, would be useless and even troublesome in common life. It is well if the mass of mankind will obey the laws when made.”⁵ Once individual rights are acknowledged, contractual negotiations become possible, and, as economists, we launch off into the interesting problems posed by the contracting or exchange process itself. We sometimes fail to recognize, or otherwise forget, that the whole institution of contract, whether this takes the

4. The “theory of property” that is implicit in my discussion here and elsewhere in the book can perhaps best be classified as some mixture of the “personality” and “utilitarian,” especially as the latter is represented by the discussion of David Hume. These and other useful classifications of theories of property, along with a good general discussion, are found in Frank I. Michelman, “Property, Utility, and Fairness: Comments on the Ethical Foundations of ‘Just Compensation’ Law,” *Harvard Law Review* 80 (April 1967), especially pp. 1202–13.

5. The citation is from Blackstone’s *Commentaries on the Laws of England*, 12th ed. (London: T. Cadell, 1794), 2:2–3.

simplest form of isolated two-party trade or the more complex n -person agreement, rests on the possibly shaky foundations of mutual agreement on individual rights, including agreement by an enforcing agent, a state, which must also limit its own behavior. It is perhaps time that economists begin to devote more attention to the origins of contract.

The modern contributions to the “economics of property” can only be welcomed. I refer to the contributions of Alchian, Cheung, Demsetz, Furubotn, Kessel, McKean, North, Pejovich, Thomas, and others.⁶ Nonetheless, the emphasis in this now burgeoning subdiscipline is on the way in which alternative property-right structures can modify individual and group behavior, with the orthodox criterion of economic efficiency implicitly at the base of the analysis. To this point, these contributions have not concentrated on explaining the emergence of property rights.⁷

6. A partial listing of the contributions is as follows: A. A. Alchian and R. Kessel, “Competition, Monopoly, and the Pursuit of Money,” in *Aspects of Labor Economics* (New York: National Bureau of Economic Research, 1962), pp. 157–75; S. Cheung, “Private Property Rights and Sharecropping,” *Journal of Political Economy* 76 (December 1968): 1107–22; Harold Demsetz, “The Exchange and Enforcement of Property Rights,” *Journal of Law and Economics* 7 (October 1964): 11–26; R. McKean, “Divergences between Individual and Total Cost within Government,” *American Economic Review* 54 (May 1964): 243–49; Douglass C. North and Robert Paul Thomas, *The Rise of the Western World: A New Economic History* (Cambridge: Cambridge University Press, 1973). These and other contributions are discussed by Furubotn and Pejovich in a lengthy review article, which also includes a full set of references. See Eirik Furubotn and Svetozar Pejovich, “Property Rights and Economic Theory: A Survey of Recent Literature,” *Journal of Economic Literature* 10 (December 1972): 1137–62.

7. Among the modern “property rights economists,” S. Pejovich is the only one who has attempted a generalized discussion of origins. He states explicitly that his aim is to show that “the creation . . . of property rights is endogenously determined” (p. 310). A careful reading of Pejovich’s paper suggests, however, that the endogeneity does not refer to the exchange process. Instead, individuals are motivated by private utility-maximizing norms to invest resources in defense and predation of resource stocks, behavior which is, of course, descriptive of Hobbesian anarchy. I prefer that we limit “economic explanations” of institutional change to the contractual process. In this context, as my analysis suggests, property rights may emerge from an economic calculus that prompts the negotiation of a “constitutional contract.” But this is quite different from the results emergent from the independent, utility-maximizing behavior of the parties. See S. Pejovich, “Towards an Economic Theory of the Creation and Specification of Property Rights,” *Review of Social Economy* 30 (September 1972): 309–25.

This may be illustrated with reference to Harold Demsetz's discussion of private-property rights among the Canadian Indians.⁸ In his interpretation of the history, the Indians of the Labrador peninsula accepted common usage of hunting grounds prior to the dramatic increase in beaver-skin prices caused by French traders. This price increase stimulated increased hunting effort; beaver skins became relatively scarce, and individual hunting tended to produce an uneconomic utilization of the common resource. As a means of internalizing the external diseconomies that this rights arrangement fostered, the Indian tribes shifted from a common-usage to a private-property structure. The historical accuracy or inaccuracy of this account need not concern us here. But note that Demsetz essentially "explains" a change in rights structure by resort to a new contractual arrangement made desirable by exogenous changes in economic data. He is employing the historical example to demonstrate the proposition or principle that there will always be a tendency for characteristics of the rights structure to be shifted toward that set which is most efficient under the conditions faced by the community. There can be no quarrel with this, and Demsetz's contribution can be acknowledged. We should not, however, make the mistake of saying that this approach explains the origin or emergence of rights among individuals or families (tribes) independent of contractual agreement, whether this be explicit or implicit. In his conceptual model, the rights of the several participants must have been mutually recognized by all participants *before* further contractual negotiations could be undertaken to change the structural characteristics.

With regard to the emergence of initial rights, we must acknowledge that any attempted economic explanation will be insufficient. How might the rights of individuals (families, tribes), rights to do things, including rights over domain, rights that are mutually respected, arise? What is the logical basis of property? To such a question, there cannot be an exclusively contractarian answer. The concept of externality may, however, prove helpful in pointing toward meaningful analysis. Demsetz argued that property rights change for the objective of "internalizing externalities." His particular discussion was directed, as noted, toward explaining how structures are shifted

8. Harold Demsetz, "Toward a Theory of Property Rights," *American Economic Review* 57 (May 1964): 347–59.

toward satisfying overall efficiency norms. He did not, at least explicitly, try to extend his thesis backward, so to speak, to examine the noncontractual elements.

This step can be taken if we introduce a sufficiently broad definition of externality. In a world without interpersonal conflict, potential or actual, there would, of course, be no need to delineate, to define, to enforce, any set of individual (family) rights, either in the ownership and use patterns of physical things or in terms of behavior with respect to other persons. I use “conflict” rather than “scarcity” here, because even if all “goods” that might be “economic” should be available in superabundance, conflict among persons might still arise. Social strife might arise in paradise. Total absence of conflict would seem to be possible only in a setting where individuals are wholly isolated one from another, or in a social setting where no goods are scarce and where all persons agree on the precise set of behavioral norms to be adopted and followed by everyone. In any world that we can imagine, potential interpersonal conflict will be present, and, hence, the need to define and enforce individual rights will exist.

The “Natural Distribution”

Consider a simple two-person world. All “goods” save one, which we shall call x , are available to each person (A and B) in superabundance. But good x is “scarce.” No production is required for its enjoyment, however, and quantities of this good simply “fall down” in fixed proportions onto each of the two persons at the onset of each period of consumption. There are no property rights, no law, in this economy. Hence, we may say that the consumption or use of a unit of x by individual A imposes an “external diseconomy” on B, and similarly, that the consumption of a unit by B imposes a diseconomy on A. In an orthodox externality setting, and notably in small-number groups where transactions costs are not prohibitively high, we might predict that “trade” between the two persons would take place, accomplishing an internalization of the reciprocal externality relationship. In our example, however, regardless of the initial distribution of x , there is no surplus to be secured through a direct trading process. Nonetheless, each of the two persons will have an incentive to “internalize the externality” that the other imposes on him.

Both A and B might seek to consume all of the good x that is available. As Thomas Hobbes perceptively noted, in this state of nature each person has a “right” to everything. Each would find it advantageous to invest effort, a “bad,” in order to secure the good x . Physical strength, cajolery, stealth—all these and other personal qualities might determine the relative abilities of the individuals to secure and protect for themselves quantities of x , which may be quite different from the relative quantities that were arbitrarily assigned by the initial disposition. These attributes may, but need not, include some inherent “live and let live” attitude among members of the common species. In any case, as a result of the actual or potential conflict over the relative proportions of x to be finally consumed, some “natural distribution” will come to be established.⁹ This cannot properly be classified as a structure of *rights*, since no formal agreement is made, although there might well exist mutual recognition of the appropriate bounds on individual action. Nonetheless, the natural distribution may represent a conceptual equilibrium, in which each person extends his own behavior in securing (defending) shares in x to the limit where marginal benefits from further effort are equal to the marginal costs that such effort requires. In the “natural distribution,” the two persons, A and B in our example, continue to impose external diseconomies on each other in the sense noted above. But now, because this distribution does offer a base from which some predictions may be made, indirect “trades” become possible that will internalize the external diseconomies.

The “natural distribution,” secured upon investment of effort in attack and/or defense of consumption shares in x , serves to establish an identification, a definition, of the individual persons *from which* contractual agreements become possible. Absent such a starting point, there is simply no way of initiating meaningful contracts, actually or conceptually. But, in the natural distribution, both A and B will recognize, on rational observation, that

9. The formal characteristics of this “natural distribution” have been worked out by Winston Bush. See his “Individual Welfare in Anarchy,” in *Explorations in the Theory of Anarchy*, ed. Gordon Tullock (Blacksburg, Virginia: Center for Study of Public Choice, 1972), pp. 5–18.

The notion of such a distribution is, of course, to be found in the work of several of the philosophers who have thought about the origin of property, notably in Aegidius, Hugo Grotius, and Thomas Hobbes.

much of the effort expended in securing and defending stocks of x is wasteful. Whatever might be the characteristics of this distribution, whether rough symmetry prevails or whether one participant becomes a consumption giant and the other a pygmy, and even if all of x is secured by one party, *both* parties will be made better off if agreement can be reached. Trades can be arranged in the sense of agreement on a set of behavioral limits. Mutual gains are possible in this way over a wide range of assignments of final consumption, with the particular assignment finally negotiated dependent on bargaining skills and other factors.

The Emergence of Property

It is appropriate to call this a genuine basis for the emergence of property rights.¹⁰ Both parties agree to and accept the assignment, which carries with it the complementary agreement that they will not behave so as to violate the terms. Both parties can, therefore, reduce their private investment in attack and defense; in the limit, the full value of x can be realized without cost.¹¹ The agreement on rights of the two parties represents a contractual internalization of an externality relationship that existed in the precontract state of nature. Note, however, that the reversion to some sort of “natural distribution” is required in order that potential trading participants can themselves be identified. This may be illustrated by tracing an alternative conjectural history of the emergence of private property among the various Indian tribes of the Labrador peninsula. The increase in the demand for beaver converted what had previously been an abundant resource into one that was scarce. No property rights previously existed, and the new scarcity produced conflict among the separate tribes. As a result of actual or potential intertribal wars, some “natural distribution” emerged that came to be recognized by all tribes.

10. This conceptual explanation is variously presented by many of the social philosophers, notably those in the general contractarian tradition. Particular versions are to be found in the works of Aegidius, Bodin, Grotius, Hobbes, and Hume. For a good summary treatment, see Richard Schlatter, *Private Property*.

11. The more general theme that all investment that takes the form of protecting rights must be, in the net, socially wasteful is developed at some length by Gordon Tullock in his forthcoming book, *The Social Dilemma*.

A reassignment of territorial hunting grounds could then have taken place by mutual agreement, with each tribe finding it advantageous because of the allowable reductions in military effort.¹²

The specific distribution of rights that comes in the initial leap from anarchy is directly linked to the relative commands over goods and the relative freedom of behavior enjoyed by the separate persons in the previously existing natural state. This is a necessary consequence of contractual agreement. In Hobbes's model, there are, by inference, considerable differences among separate persons in a precontract setting. To the extent that such differences exist, postcontract inequality in property and in human rights must be predicted. For my purposes, there is no need to discuss in detail the degree of possible inequality among separate persons in the conceptual state of nature that has been used to derive the logical origin of property rights. Those who have referred to the strong enslaving the weak may well have exaggerated the differences. The romantic moderns, on the other hand, who adopt their own variants on Rousseau's noble savage may be equally off the mark in the opposing direction. To make meaningful statements here, it would be necessary to sift and to examine the available anthropological, ethological, and historical evidence, a task that lies beyond both my competence and my interest.

Nor need the analysis depend critically on the acceptance or rejection of any particular model or hypothesis about human behavior. We need not follow Hobbes and assume that men behave from narrowly defined self-interest. We could assume, equally well, that even in some state of nature men behave in accordance with self-interest tempered by regard for their fellows. Or, in the other limit, we might also assume that individuals adopt precepts for behavior that reflect the interest of the human species. The inequalities among persons that may be conceptually observed in the "natural distribution" will result both from the inherent differences in personal capacities and in the types of behavior actually adopted. If, for example, personal capacities should be widely different, but, at the same time, all persons behave nonindividually, the observed natural distribution might reflect considerably less in-

12. For a paper that specifically introduces the Demsetz example, but which interprets the emergence of property rights in a manner fully consistent with that which is developed here, see Charles R. Plott and Robert A. Meyer, "The Technology of Public Goods, Externalities, and the Exclusion Principle," Social Science Working Paper No. 15 (revised) (California Institute of Technology, February 1973).

equality than that which would be observed under hedonistic behavior patterns. Since all we can observe, even conceptually, is the natural distribution itself, the precise combination of inherent personal qualities and behavioral norms is not relevant. It is, however, relevant to note that there is no basis for assuming equality in the conceptually observed distribution among persons in the natural state, and, consequent on this, no basis for predicting equality in the agreed-on allocation or distribution of rights in the initial post-contract setting. There is nothing to suggest that men must enter the initial negotiating process as equals. Men enter as they are in some natural state, and this may embody significant differences.

Violations of Contract

The gains-from-trade that are potentially achievable by an agreement on rights are realized by all parties through the disinvestment in socially wasteful effort devoted to both predatory and defense activity. An agreed-on assignment will not normally be stable in one particular sense. Once reached, one or all parties may find it advantageous to renege on or to violate the terms of contract. This applies to *any* assignment that might be made; the tendency toward individual violation is not characteristic of only some subset of possible agreements. Within the setting of an agreed-on assignment of rights, the participants in social interaction find themselves in a genuine dilemma, familiarized under the “prisoners’ dilemma” rubric in modern game theory. All persons will find their utility increased if all abide by the “law,” as established. But for each person, there will be an advantage in breaking the law, in failing to respect the behavioral limits laid down in the contract.

This may be illustrated for our two-person model through a simple matrix, as shown in figure 2.1. In the natural distribution, in which neither party acknowledges or respects any rights to the single scarce good, the utility levels achieved are shown in Cell IV. The left-hand numbers in each cell represent utility indicators or net payoff values for A, the right-hand numbers those for B. As indicated, the utility payoffs need not be equivalent in this anarchistic “equilibrium.” (We need not introduce complexities about interpersonal utility comparability at this point since it is the absence of equality that is important, not the demonstration of inequality.) Upon agreement, the utility payoffs are those shown in Cell I, where both persons are in positions

		B	
		Respects Rights	Respects No Rights
A	Respects Rights	Cell I 19, 7	Cell II 3, 11
	Respects No Rights	Cell III 22, 1	Cell IV 9, 2

Figure 2.1

that are superior to those achieved in Cell IV. Also, joint or combined payoffs are maximized in Cell I. (Here, again, qualifications concerning interpersonal comparability and, hence, additivity would be required for completeness.) As depicted, however, both A and B have private incentives to renege on or to violate the contractual agreement on rights provided that they can do so unilaterally. The position for individual A is improved if he can secure a shift to Cell III, while that for individual B is improved if he can somehow get to Cell II.

In the strict two-person setting, either one or both of the persons may refrain from violating the contract because he may rationally anticipate that the other's reaction would force a quick return to the precontract state of nature. In the simple game setting of figure 2.1, Cell I is in the "core," to introduce a term specific to modern game theory. No player can guarantee for himself a better outcome than that which he secures under universal adherence to the agreed-on rights assignment. No person can assure himself that breaking the law will not result in a worsening of his own position. This characteristic suggests that there are important elements of stability in a Cell I position. Such stability tends to disappear, however, as the number of participants in the interaction increases, even if the formal properties remain unmodified. Furthermore, variations in the characteristics and relationships within the payoff structures, which will be introduced in later chapters, may eliminate the stability that here is apparent.

Two-Stage Contract

To this point, the conceptual emergence of some initial assignment or imputation of individual rights has been discussed in an extremely abstract and

simplified model. I have assumed that only one scarce good exists other than time itself. The model is also limited to a two-person interaction; and it is timeless, which amounts to saying that none of the elements change over time. It will be necessary to relax each of these restrictions. The treatment must be generalized to allow for scarcity in many goods, to explain the interaction of many persons; and the effects of time in modifying the contractual relationship must be incorporated directly into the analysis. In the remainder of Chapter 2, I shall make the first two extensions. The introduction of time is sufficiently complex to require additional space.

We may continue to use the two-person model but allow for more than a single scarce good. Conflict between A and B in the state of nature will now arise over the disposition of all of those goods that are not superabundant. For present purposes, we shall continue to assume that these goods are not produced, but instead that quantities of these goods simply “fall down” in some distribution between the two persons.

As in the one-good model, a natural distribution will emerge from actual or potential conflict. This distribution, which will now be many-dimensional, will be influenced by the initial disposition of the goods, and by the relative personal characteristics and behavior patterns of each person. The outcome may be described by a vector whose components represent net amounts of each of the scarce goods finally enjoyed in consumption by each person,

$$(X_A, X_B; Y_A, Y_B; \dots).$$

In attaining his share in this natural distribution, each person finds it necessary to invest effort (time and energy) in predatory and/or defense activity. There is no difference in this respect between this and the one-good model. And, as before, the natural distribution provides the base from which contractual agreements become possible. Agreement will, as before, take the form of some mutual acknowledgment of rights. Gains are secured from the reductions in predation-defense effort. This contract, which becomes the initial leap from Hobbesian anarchy, is the first stage of a two-stage contractual process. For purposes of convenience, I shall, here and later, refer to this as “constitutional contract.”¹³

13. The position taken here is that *both* constitutional and postconstitutional rights conceptually emerge from *contract*, but that it is essential that the two stages be kept distinct. This may be compared with the position advanced by F. A. Hayek in his work *Law*,

It is the existence of two stages or levels of agreement that distinguishes the many-good from the one-good model. In the latter, the initial agreement on shares in the single scarce good represents the limit to trade. No further contracts between the two persons will offer mutual gains. The two-person group attains the Pareto-frontier through its initial agreement on rights of disposition. The many-goods model differs sharply at precisely this point. If individual tastes differ, there may be potential gains-from-trade over and beyond the initial "trade" of agreements on individual rights. The trading process at this second, or *postconstitutional*, stage is, of course, the domain of traditional economic theory. Individual participants are assumed to enter the potential trading arena with identifiable endowments and/or capacities, and their rights to these initially held endowments are assumed to be mutually accepted by all members of the community and to be enforced by the state. The two-stage contractual sequence may be depicted simply with a diagram like that of figure 2.2. The utility attained by individual A is shown on the ordinate, that attained by individual B on the abscissa. The natural distribution, that outcome which emerges as a quasi-equilibrium in the genuine state of nature, is shown at D. The initial constitutional contract, which involves nothing more than mutual agreement on some structure of rights, shifts the outcome in a general northeasterly direction, bounded as indicated by the dotted lines drawn from D. Suppose that the actual agreement shifts the utility positions to C. In the one-good model, no further trades are possible, and the utility-possibility frontier is attained for the two-man community. In the many-goods model, however, further trades in specific goods may be mutually advantageous. This will always be possible if tastes differ and if the agreed-on assignments do not correspond precisely to the preferred final-goods packages.

There seems little reason to predict such precise correspondence, although

Legislation, and Liberty, vol. 1, *Rules and Order* (Chicago: University of Chicago Press, 1973). If I interpret his argument correctly, Hayek suggests that "law," which is equivalent to what I have called constitutional contract, is not contractual in origin but emerges from an unpredictable evolutionary process. He advances this argument in opposition to the "constructivists," who are alleged to think of law as willed by someone. In historical fact, evolutionary elements may explain much of the emergence and development of "law." Acceptance of this does not, however, negate the application of contractual-constructivist criteria in an evaluation of that "law" which exists, and which might be willfully modified.

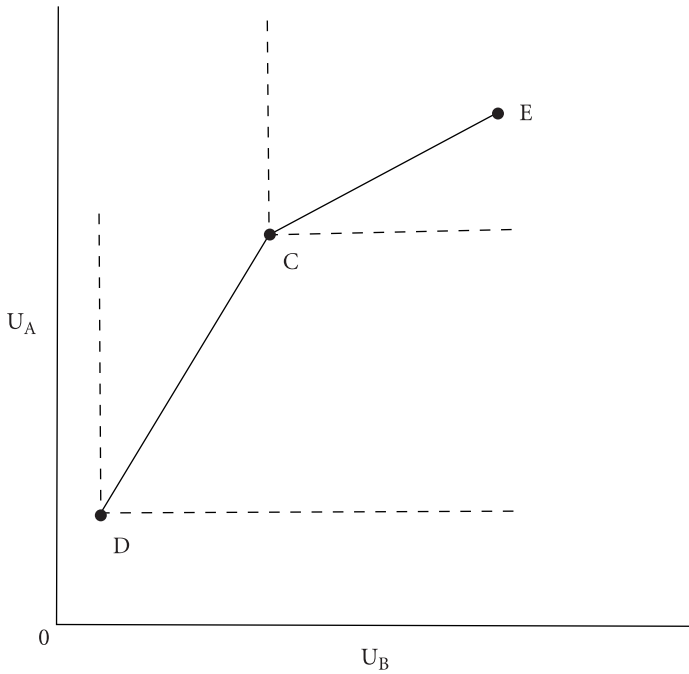


Figure 2.2

some tendency toward correspondence might be present. In the natural state, individuals will devote more effort to securing and protecting those final goods that stand relatively high in their own preference rankings. Furthermore, in the conceptual bargaining over the rights to be accepted, individuals will tend to sacrifice goods that they value relatively lower than goods which they secure in “exchange.” Relative abilities to secure and defend goods, and to secure goods in conceptual bargaining, need not correspond closely with relative values placed on goods by the separate participants. Under any combination of these elements that generates an absence of precise correspondence between originally agreed endowments and optimally preferred bundles, trades can be made, and these trades will move the utility positions further in the northeasterly direction, with the bounds for postconstitutional trades indicated by the dotted lines extending from C. In some final trading equilibrium, say, E, the utility-possibility frontier is attained. The constitutional contract involves the initial shift from D to C; postconstitutional con-

tract involves the shift from C to E. Economists have concerned themselves almost exclusively with the latter to the neglect of the former.¹⁴

Further classification that is more familiar to economists can be made within postconstitutional contracts or exchanges. The shift from C to E can take place through two types of process. In the first, individuals exchange goods that are rival in usage and fully partitionable or divisible among persons, both at the stage of initial rights assignments and at the stage of final consumption. If X and Y are two such goods, and individual A is initially given rights to relatively more X, he will then give up units to B in exchange for units of Y. This trade, which we may call *private-goods* trade, using conventional terminology, may be institutionalized through markets when large numbers of potential traders are interlinked.

Mutual gains may continue to exist, however, over and beyond these achievable by trades in purely partitionable goods. Joint rather than separate consumption of some goods may be relatively efficient. To illustrate, let us suppose that both goods X and Y are rival in consumption of A and B, but that good Z is potentially able to meet the demands of both persons simultaneously; there is no rivalry in consumption with respect to Z. We continue to assume that we are examining a no-production economy, with initial supplies of goods arbitrarily distributed. In this setting, some question might be raised as to why Z, defined as a purely public good, again to use conventional terminology, should be included in the “scarce” category with individual rights to its disposition assigned in an initial contract at the constitutional stage. If treated independent of all other goods, there would be no need for any such assignment; indeed one means of describing a purely collective or public good is to say that no distributional problem exists.¹⁵ But when distributional conflict characterizes other private or partitionable goods, X and Y in this example, individuals may seek to establish rights over the distribution of Z as an indirect means of securing more X and Y in the postconstitutional trading process. This will become more relevant as we move to more

14. Note that the distinction made here need not require that any historical relevance be attributed to constitutional contract. No matter how a structure of property rights may emerge, it is useful to separate the definition of structure from the exchange of rights within this structure.

15. For an elaboration of this point, see my *Demand and Supply of Public Goods* (Chicago: Rand McNally, 1968), especially chap. 9.

complex and more realistic models later in the book. In any case, we assume here that the agreed-on assignment of rights provides net quantities of both private and potentially public goods to A and B. For illustration, suppose that this structure assigns all units of X to A, and all units of both Y and Z to B. Trade in private goods, X and Y, proceeds in the orthodox manner. For the good that is nonrival in final use, Z, individual B will agree to make some (all) of it available jointly to A upon the payment, by A, of some appropriately negotiated quantity of X, which is rival in final usage.¹⁶ This sort of “trade” with respect to Z is different in kind from trade among persons in ordinary goods and services that are rival in consumption. The shift of a community from the position depicted by the movement from C to E in figure 2.2 is accomplished by some combination of *private-goods trade* and *public-goods trade*, both of which take place within the limits defined by constitutional contract.

From Small to Large Numbers

Before proceeding from the two-person to the n -person model, it is perhaps useful to summarize the conceptual schemata that has been developed. From a

1. natural distribution, a
2. constitutional contract is negotiated, from which, in turn,
3. postconstitutional contract becomes possible, through
 - a. private-goods trade (goods rival in consumption) and/or
 - b. public-goods trade (goods nonrival in consumption).

When we introduce a large number of participants, negotiations may take place among subgroups or coalitions contained within the larger and more inclusive community membership. A “natural distribution” may, therefore, be conceived at any one of several levels of aggregation. At one extreme, which we might call the pure natural distribution, no coalitions exist, and

16. In the elementary discussion here, I am assuming that B is able to exclude A from the use of Z without excessive cost. If nonexclusion is inherent in Z, then no property right will be initially assigned in this good in the no-production setting. If Z must be produced, however, the discussion becomes applicable, even if, once produced, Z is nonexcludable. Exclusion will, in this case, take the form of not producing.

each person acts strictly on his own in the genuine Hobbesian “warre” of each against all. From this base, however, “constitutional contracts” may be made among members of groups of any two or more persons, with internal assignments of rights, while as among the separate groups conflict continues. At such levels, the separate subgroups or coalitions (some of which may include only one member) are in a natural distribution fully analogous to that described for the two-person model. The process of contractual internalization may proceed as the subgroups become larger until some final negotiating process which incorporates all persons in the community within a single constitutional structure.

The final or ultimate constitutional contract will define the rights assigned to each person in the inclusive community. And each person will find his own position improved over that which he might have enjoyed in any one of the natural distributions noted above, because he will not have to exert or contribute effort to defense and predation, either as an individual on his own account or as a contributing member of a subset of the total community.¹⁷ The second, or postconstitutional, stage of negotiation can commence. At this point, the shift from the two-person to the n -person model introduces other important differences in results. With respect to private-goods trade, exchanges can proceed in the orthodox manner that has been exhaustively

17. Complexities arise in the large-number setting which make any assignment of rights less stable than in the small-number setting. There are two reasons for this difference. In the first place, almost any imputation or assignment that is agreed on will dominate that which the individual would anticipate were he to opt out and to try to exist on his own in pure anarchy. With large-number communities, however, the set of rights assignments that will dominate, for each person, the position that he might expect to secure if *any* coalition opts out, is much more restricted, and, indeed, this set might be empty in many interactions. These aspects of interactions have been discussed in detail, formally, in modern game theory. To use the appropriate terminology, for large-number groups there may be no imputation or assignment that is in the core, and, if a core does exist, the number of imputations contained may be small. For an introductory discussion of these concepts, see Duncan Luce and Howard Raiffa, *Games and Decisions* (New York: John Wiley and Sons, 1957).

Second, even if the assignment finally arrived at should qualify for inclusion in the core in the game-theoretic sense, individuals would still find it advantageous to violate the terms if they predict an absence of response on the part of remaining members of the group. This tendency toward instability is present in both the small- and large-number groupings, but the sheer impersonality or anonymity of individuals in large groups makes strategic behavioral calculations much less likely to occur.

treated by economists. Market institutions will be formed; bargaining ranges will be restricted by the presence of multiple alternatives for each buyer and each seller of goods; outcomes will tend to be more determinate than in the small-number setting. The important point to be stressed, for my purposes, is that these exchanges in private goods still take place between separate pairs of traders, between individual buyers and individual sellers. Each exchange remains a two-party transaction, as in the simplified model, despite the addition of numbers. There is no necessity to bring all members of the community into each contract. There is nothing that might be classified as “social contract” here.

Over and beyond all such exchanges in private goods, there may exist further potentially realizable surpluses from “trades” in public or collective-consumption goods and services, those that can simultaneously meet the demands of *all* persons in the group. In reference to our simple example, suppose, as before, that Z is such a public good, which may be consumed jointly and simultaneously by all members of, say, a three-man community, A, B, and C. Assume, as before, that all of the available supply of Z is under the disposition of individual B under terms of the basic constitutional contract. “Trades” can be made which involve a transfer of privately partitionable goods from A and C to B in exchange for the latter’s willingness to make available Z for joint consumption. The point to be emphasized here is that, in such trade, *all* members of the final consuming group must be brought directly into the contractual negotiations with respect to provision of the public good. It is not possible to factor down a network of economic exchange into separate two-party transactions. This feature categorically distinguishes public-goods trade from private-goods trade when large numbers are involved. With the former, with public-goods trade, something akin to “social contract” again comes into being, comparable in numbers of participants to the constitutional contract that delineates individual rights.

Before we can develop further the implications of this similarity, and the confusion that the similarity fosters, certain ambiguities must be clarified. For expository purposes, I have assumed that all scarce goods fall into either one or the other polar extreme—the purely private good or the purely public good.¹⁸ Furthermore, I have implicitly assumed that the “range of public-

18. This is the same assumption that was made by Samuelson in his classic formulation of the welfare norms for public-goods provision. See Paul A. Samuelson, “The Pure The-

ness" for all goods in the public-goods classification extends precisely to the limits of the inclusive community membership. This assumption is perhaps even more restrictive than the polar classification. There may exist many goods that qualify as "public" or "collective" over differing numbers of consumers but which, at the same time, become fully rival in consumption as the number of users is increased. The size of the consuming "club" that meets efficiency criteria may be much less than the total membership of the community.¹⁹ If such "clubs" are small, relative to the size of the total community, market institutions can emerge that will internalize the required contractual arrangements, even though the strict pairwise exchange-characteristics of pure markets are violated. For our purposes, having recognized that much of the noncollective sector may embody such small-number jointness, we can treat all "trades" that do not involve substantial fractions of the total community membership as taking place in "private goods." In somewhat more practical terms, this amounts to a neglect of local government decision-making, as such.

The analysis suggests that "social contracting," defined as those negotiations which involve *all* members of the community, may take place conceptually at two levels or tiers: at some initial stage of constitutional contract, in which agreement is reached on an assignment of individual rights, and at some postconstitutional stage in which individuals agree on quantities and cost shares of jointly consumed goods and services. Essentially the same problems emerge at each tier, problems that are created largely by the necessity to bring large numbers of persons into the same contractual arrangements. The sheer cost of getting agreement on any outcome rises sharply as numbers increase. In two-party trading, only the single buyer and the single seller need agree. This difference in numbers alone makes for major differences in costs. But the absence of alternatives exacerbates the attainment of agreement in many-party negotiations. It may not prove overwhelmingly costly for very large numbers of persons to reach agreement if each participant has available to him other avenues for securing comparable objectives. To the extent that

ory of Public Expenditure," *Review of Economics and Statistics* 36 (November 1954): 387–89.

19. For a more extended treatment, see my "An Economic Theory of Clubs," *Economica* 32 (February 1965): 1–14.

an individual has other opportunities, he can withdraw from participation in the large-number group if a proposed settlement does not fall within his broadly defined preferences. For the single inclusive community, however, the necessary participation of all members eliminates the prospects of effective alternatives. National states, from which out-migration can take place normally only at significant costs to an individual, rather than local governments, become the real-world organizational units to which the analysis is most directly applicable.

A final qualification must be introduced at this point in order not to give the appearance of inconsistency with the discussion in subsequent chapters. This involves the delineation or assignment of individual rights emerging from constitutional contract. It is essential that the potential variability in the specification of individuals' rights be recognized. The conceptual agreement may range from an assignment that invokes relatively little "law" in the formalized sense to an assignment that rigidly constrains individual behavior over many dimensions of adjustment. An assignment of rights is not an all-or-nothing choice. *Ceteris paribus*, the liberty inherent in anarchistic order without law is the most desirable state of affairs. The extent to which an individual, or the community of individuals, may be willing to trade off the liberty that remains present even in the Hobbesian jungle for the stability promised in regimes with varying degrees of formal restrictiveness will depend on the nastiness of the jungle, the value placed on order, the costs of enforcement, and on many other factors, some of which will be subsequently discussed.

3. Postconstitutional Contract

The Theory of Public Goods

In this chapter I shall discuss in more detail the 3b category in the conceptual schemata. This category includes the many-party agreements, the genuine “social contracts,” that may take place after (1) individual rights are assigned in constitutional contract, and (2) all gains-from-trade in strictly private or partitionable goods are realized. The analysis may be summarized under the rubric “theory of public goods” defined in an inclusive sense. The basic analytical material is familiar to modern economists, but the setting for the discussion may be sufficiently novel as to warrant further consideration.

I have modified the logical order suggested by the schemata itself. This would place conceptual constitutional contract which embodies the definition and assignment of individual rights ahead of and prior to postconstitutional contract, which embodies trade among persons in both private and public goods after rights have been defined and mutually accepted. I have reversed the order of analysis for didactic purposes. The theory of postconstitutional contracting and exchanging, whether this be the theory of trade in private goods (the domain of orthodox microeconomic theory) or the theory of trade in public goods, is less complex than that of constitutional contract. Indeed it is largely because the complexities of the latter are ignored or assumed away that we are able to develop sophisticated and rigorous analyses of the second set of personal interactions.

Except for an introductory paragraph in the following section, I am also neglecting the most important category of postconstitutional contract itself, the exchange of private or partitionable goods and services. In part this omission stems from the familiarity of this theory. But, more importantly, this apparent gap in the treatment is justified by my larger purpose in this book. The orthodox theory of exchange in private goods is not itself the source of

major confusion about the role of the state. The same cannot be said of the complementary theory of public goods.

As an additional limitation, the emphasis will be on postconstitutional "social contracting" in its purest form. I shall not treat the interesting analytical and practical issues that arise when consumption jointness and/or nonexclusion efficiencies suggest many-party bargains over groups that are smaller than the membership of the total community, a membership which is assumed to be set exogenously.¹ That is to say, neither fiscal federalism nor the theory of clubs will be examined.² Discussion of these issues, intrinsically interesting as they are, would distract attention from those problems in many-person social contract which I want to discuss in this book.

In this chapter I shall continue to work within what is essentially a timeless model; contracts are assumed to be immediately carried out, and by the same persons who enter the agreement.

Market Failure and the Free-Rider Problem

If individual rights are well defined and mutually accepted by all parties, persons will be motivated voluntarily to initiate trades in partitionable goods and services, those that are characterized by full or quasi-full divisibility among separate persons or small groups. That is to say, markets will emerge more or less spontaneously out of the self-interested behavior of individuals, and the results will be beneficial to all members of the community. The potential gains-from-trade will be fully exploited, and all persons will be better off than they would have been by remaining in their initial postconstitutional positions, with well-defined endowments and capacities imbedded in a structure of legally binding human and property rights. The genius of the eighteenth-century moral philosophers (notably Mandeville, Hume, and

1. For a paper which discusses some of the problems which this neglect creates, see Dennis Mueller, "Achieving a Just Polity," *American Economic Review* 44 (May 1974): 147–52. Although Mueller develops his argument in the context of a critique of the work of John Rawls, much of his analysis is directly relevant to my argument in this book.

2. For two recent books which examine many of these issues in detail, see Wallace E. Oates, *Fiscal Federalism* (New York: Harcourt Brace Jovanovich, 1972), and Richard E. Wagner, *The Fiscal Organization of American Federalism* (Chicago: Markham Publishing Co., 1971).

Smith) lay in their discovery and application of this simple principle, which has been variously elaborated in modern economic theory, the principle which, directly or indirectly, served as the basis for organizing the institutions responsible for post-Enlightenment economic progress in the Western world.

A critical feature of the spontaneous and efficient order of markets is the two-party contractual setting which serves to reduce agreement or transactions costs to minimal levels. Trades are consummated when terms are established, and only two persons need agree explicitly. Furthermore, the fact that two-party exchanges are linked in a network of alternative options facilitates rather than retards agreement on terms. Increased numbers multiply the potential alternatives available to individual buyers and sellers and narrow the ranges of dispute inside particular exchanges and between specific trading partners.³

The thrust of the modern theory of public or collective-consumption goods is the demonstration that markets fail to emerge and to produce tolerably efficient results when potential contracts require the simultaneous agreement of many parties. Neither of the efficiency-generating elements of private-goods markets is present in the pure public-goods model. Agreement or transactions costs are much higher because of the large number of persons who must be brought into the same bargain or exchange. And this inclusiveness itself tends to eliminate potential alternatives for participants, alternatives which narrow the range over which terms of trade might settle. The basic behavioral contrast between private-goods exchange and public-goods exchange is often pointed up by reference to the “free-rider problem” in the latter, although this terminology is itself somewhat misleading.

In a simple two-party trade in divisible goods, each participant is aware that the behavior of his trading partner is directly dependent on his own action. If you have oranges and I have apples, and I want some of your oranges, I know that my desires can be met only by giving up apples, by paying some price. I can scarcely expect you to offer me oranges independent of my own behavior, nor could I expect to pick up the oranges without you invoking

3. For a discussion of the spontaneous order that emerges from market processes, see the excellent essay by Michael Polanyi, *The Logic of Liberty* (Chicago: University of Chicago Press, 1951).

rights of proprietorship. If I try this, or try to renege on a contract that I have made, I cannot expect to escape the costs that result from a breakdown in exchange. There is no way I can expect to get the oranges without cost. The behavioral setting for exchange or contract in indivisible or public goods is dramatically different. Suppose that many persons (1, 2, . . . , n) want something done, some “good” that is completely nonrival in usage. David Hume’s example of the drainage of the village meadow is illustrative.⁴ Suppose that each of the many villagers knows that the drainage would prove beneficial to him personally if the costs were shared equally among all members of the group. Even more desirable for an individual, however, would be the situation in which the meadow is drained by others, allowing the individual to escape without making any contribution. Each person will be motivated to refrain from voluntary initiation of action here to the extent that he expects his own behavior to be independent of that of other participants in the potential social interaction.

Each person has an incentive, therefore, to try to become a “free rider,” one who secures the benefits of the jointly consumed good or service without participating fully in the sharing of its costs. As noted above, this “free-rider” terminology is somewhat misleading in that it suggests strategic behavior on the part of the individual participant. Strategic behavior designed to conceal from others the individual’s true preferences for the public good will take place, however, only if the group is itself small and if the individual recognizes that his own behavior can affect others. With large numbers, the behavioral setting is quite different, although the results are similar. Here the individual participant does not behave strategically vis-à-vis his fellows; he treats their behavior as a part of his environment, and he does not consider that his own action can exert any influence on that of others in the sharing group. In this setting, the individual maximizes his utility by refraining from making an independent contribution toward the provision and the financing of the commonly shared good or service. In either case, some of the potential gains-from-trade that are available to all members of the group will not emerge spontaneously, even if individual initial rights are well defined and

4. David Hume, *A Treatise of Human Nature*, p. 538. The behavioral principle involved here has been recognized for centuries, or at least since Aristotle. See Aristotle, *Politics*, trans. H. Rackham (Cambridge: Harvard University Press, 1967), p. 77.

enforced. Exchanges in genuinely public goods will not be consummated voluntarily in the same institutional framework that facilitates exchanges in private goods.⁵

Exchange and Unanimity

Ordinary exchanges in private goods can be described as taking place under implicit unanimity. That is to say, if a buyer and a seller agree on terms, an exchange takes place and all members of the community outside this two-party relationship acquiesce in the outcome. Explicit agreement is not required on the part of these outsiders, and if any of this group should have desired to interfere with any observed exchange, he had the option of offering more favorable terms to either buyer or seller. So long as the spillover or external effects produced by the exchange are not significant, two-party trading under such implicit unanimity satisfies criteria for efficiency.⁶ If, however, the characteristics of the goods are such that all members must participate explicitly in efficient sharing arrangements, the unanimity required becomes much more formidable. All persons must explicitly agree on the trading terms. Efficiency may require that the whole collectivity of persons be organized as an inclusive unit for implementing public-goods exchanges.

Knut Wicksell was the first scholar to recognize that a rule of unanimity for reaching collective decisions provides the institutional analogue to two-person trade in strictly private or partitionable goods.⁷ The inclusive coalition of traders that is required to exploit fully all potential surplus will not, however, emerge naturally or spontaneously from the private, utility-maximizing behavior of persons who find themselves in a pure public-goods interaction. This remains true even if we ignore the agreement-transactions costs reflected

5. For a more detailed discussion of the basic analysis here, see my *Demand and Supply of Public Goods*, chap. 5.

6. For a summary discussion, see Ludwig von Mises, *Human Action: A Treatise on Economics* (New Haven: Yale University Press, 1949), p. 271.

7. See Knut Wicksell, *Finanztheoretische Untersuchungen* (Jena: Gustav Fischer, 1896). Major portions of this work are available in translation under the title "A New Principle of Just Taxation," in *Classics in the Theory of Public Finance*, ed. R. A. Musgrave and A. T. Peacock (London: Macmillan, 1958).

in higgling over terms of trade. A “natural equilibrium” in which some coalition of persons provides some of the public good and in which some other members of the community remain outside as free riders may emerge spontaneously in particular instances, but these results will tend to be inefficient. Careful analysis suggests that if efficiency criteria are to be met, some “social contract” among *all* persons must be made, a contract that requires all members of the community to participate in collective decisions which are, in turn, made under a unanimity rule. There is an apparent paradox here worth noting. A rule of unanimity will insure to each individual that he will not be harmed or damaged by collective action. But individuals, until and unless they are specifically organized under a “social contract” like that indicated, will not, privately and independently, attain efficient outcomes through voluntary trades or exchanges.

This poses a question of some importance for my discussion. Does a “social contract” in which all members of the community agree to make all collective choices relating to the provision and cost-sharing of a purely public good embody coercion as meaningfully defined? Ex ante, each participant knows that he will secure gains under such a contract, gains over and beyond those secured when none of the pure public good is provided. Recalcitrant members of the community may, however, expect to be able to secure differentially larger gains by remaining outside of the possible cost-sharing coalitions that would emerge to provide some of the jointly consumed good. Since, by our assumptions, Pareto optimality or efficiency is not attained until all persons are brought into the trading arrangements, there must exist mutual gains-from-trade as between potential cost-sharers and any potential free riders in this “social contract” sense. Hence, it would seem that an agreement to join a collectivity that would make its decisions only under a rule of unanimity could be reached noncoercively. Such an agreement might require, however, that certain members of the group be allowed differentially higher gains solely because of their unwillingness to cooperate. On the other hand, if this sort of differential treatment is granted, it might, in its turn, prove unacceptable to persons who would otherwise voluntarily agree to the contract. The basic principle of collective political order, that of equal treatment, would be violated at the outset. Paradoxical as it might seem, the conclusion must be that an all-inclusive collectivity could scarcely be organized voluntarily,

even one that is severely limited to some required adherence to a rule of unanimity in making all collective choices.⁸

Unanimity, Voluntarism, and Exclusion

This result is modified if exclusion can be implemented. If those persons who do not choose to join in collective arrangements under which all cost-sharing decisions are to be made unanimously (surely a minimal set of requirements) can be excluded from any enjoyment of the subsequent benefits of public-goods provision, Pareto optimality or efficiency will tend to be attained voluntarily even in the pure public-goods case. Sharing arrangements will tend to emerge in which all persons will participate.⁹ Potential free riders will not exist since each person will recognize that, should he refuse to participate, he will be wholly excluded from the enjoyment of public-goods benefits, benefits that he evaluates positively. Exclusion, as such, might never be observed in such a setting since the certainty of being excluded if they stayed outside would motivate all persons to join in the basic contract. Exclusion, if practiced, may in some cases be extremely costly to those in the sharing group, and in the case of a good that exhibits no rivalry in consumption or use, exclusion is always resource wasteful. Nonetheless, to the extent that the power to exclude and the willingness to exclude are known to exist, the sort of inefficiency generated by free-rider behavior will rarely, if ever, occur.¹⁰

More important for my discussion, however, is the problem of reconciling

8. For a discussion of the importance of others' behavior in influencing the willingness of individuals to participate in group decisions, see William J. Baumol, *Welfare Economics and the Theory of the State*, rev. ed. (Cambridge: Harvard University Press, 1965).

9. Note that I am defining a pure public good here in terms of the jointness rather than the nonexclusion property. That is to say, purity implies that additional consumers or users of any quantity may be added at zero marginal cost to those already in the group. It need not, in this context, be prohibitively expensive to exclude individuals from enjoying the benefits.

10. Exclusion from public-goods benefits is analytically equivalent to punishment for law violation, which will be discussed in detail in Chapter 8. The possible unwillingness of members of a cost-sharing group to exclude free riders because of the costs of the exclusion is conceptually equivalent to the possible unwillingness of law-abiding members of a polity to punish offenders because of the disutility involved in the act of punishment itself. In this, and other examples, an understanding of the central theory of public goods is helpful in understanding some of the issues involved in maintaining order under law.

a power of exclusion, regardless of cost, with the assumption that individuals' rights have been assigned, rights which are well defined and are mutually accepted by all parties. If we shift to the orthodox economist's model here and specify these in strict commodity dimensions, representing property rights as individualized claims to stocks of resources and final goods, complexities arise. Suppose that, in a community of n persons, $n - 1$ of them express a willingness to enter a binding contractual agreement to participate in sharing arrangements for the provision of a purely public good, with decisions as to quantity and cost shares to be made only under a rule of unanimity. When implemented, this would involve each of the $n - 1$ persons giving up some share of his initial endowment or stock in "exchange" for the expected return flow of public-goods benefits. But what about the n^{th} person who refuses to enter the contract? We have implicitly presumed that, initially, he is a "member of the community," but we have not specified just what such membership means. Exclusion from enjoying the benefits of the public good must involve complete or partial expulsion from the community, as such. But this act seems inconsistent with the assignment of rights in basic constitutional contract, the assignment that defines an individual in terms of his initial endowments and entitlements which, presumably, incorporate physical location in a geographic community along with "social" location in a community which has adopted certain generalized rights of citizenship. Exclusion from the enjoyment of the public good may involve coercive intrusion into an individual's rights, defined independently of the public-goods decision.

There are two avenues of escape from this logical difficulty. We can either (1) reject the logical possibility of genuine social contract, even at the post-constitutional stage, while continuing to define individual rights in the manner suggested, or (2) redefine the assignment of rights in constitutional contract so as to embody exclusion. As the discussion indicates, the latter course offers a more explanatory model for exploring the complex issues of social order. Specifically, I propose that the initial assignment of individual rights emerging out of some prior constitutional contract embodies sets of personal claims to physically defined resource endowments (human and non-human) along with claims to share generalized rights of citizenship, limited or constrained by the minimal negation of such claims as may be required to implement exclusion from the benefits of public-goods provision upon the

expressed unwillingness of claimants to participate in the postconstitutional contract under an effective rule of unanimity. Put somewhat more simply, this means that membership in a community is defined so as to compel participation in the genuine postconstitutional contracting for public goods, provided that an effective rule of unanimity is insured. Compulsion takes the form of exclusion from public-goods benefits, exclusion that may, if necessary, require the negation of specific "private ownership" claims.¹¹

This setting allows us to analyze postconstitutional contract in a fully voluntaristic model. The individual, as such, is defined in terms of the rights assigned to him in constitutional contract. These include a specific imputation of initial endowments along with membership in a collective unit that makes decisions in accordance with a unanimity rule. The individual possesses no right to withdraw from the collectivity; to do so violates the constitutional contract just as clearly as the physical taking of endowments or goods that are assigned to other persons. Admittedly, this is a highly abstract and unrealistic construction, but it is necessary in order to develop the more realistic constructions to follow. The model suggests that, even when we restrict decision-making by a rule of unanimity, a collectivity, a state, must be an explicit element of and emerge out of constitutional contract. The inclusive exchanges in purely public goods that efficiency dictates will not necessarily emerge voluntarily from the behavior of individuals, each of whom is defined only in terms of initial endowments.

If, however, we make membership in an explicitly organized political entity an inherent component of each individual's rights, and if we restrict decision-making of this unit by a rule of unanimity, we can discuss all-inclusive, many-party exchange in public goods in voluntaristic terms, analogous to the two-party exchanges in private or partitionable goods implemented through market processes. As Wicksell recognized, the rule of unanimity offers the only ultimate test for efficiency in many-party exchanges, efficiency being measured by individualistic criteria. Or, to put this differently, any multiparty exchange that captures potentially realizable surplus can conceptually secure the unanimous approval of all participants. (In positive-sum

11. Mancur Olson has stressed the importance of by-product private goods as a means of enforcing exclusion in public-goods provision. See his *The Logic of Collective Action* (Cambridge: Harvard University Press, 1965).

games, all players can gain.) To get this result, however, individuals' incentives to invest in pure distributional gains must, somehow, be reduced or eliminated. A rule of unanimity provides each and every participant with a veto over final outcomes; it places each person in a position where he can bargain bilaterally with all others, treated as a unit. Because of this feature, the costs of agreement under a unanimity rule may be extremely high or even prohibitive. Recognizing this, Wicksell himself was willing to propose a qualified unanimity rule, by which he meant something like five-sixths of the total membership (or their representatives) for fiscal choice-making.

Individual Rights under Nonunanimity Rules

On grounds of institutional efficiency, departure from unanimity in the reaching of collective decisions seems necessary. Nonetheless, the importance of this change in any derivation of any postconstitutional "social contract" should be stressed. In *The Calculus of Consent*, Gordon Tullock and I analyzed the choice faced by an individual at the stage of constitutional contract on the presumption that his own cost-benefit position in subsequent decisions is unpredictable. We derived a logical basis for the adoption of less-than-unanimity rules, although we did not present arguments for any specific one among a large set of alternatives.¹² Indeed, one of our subsidiary purposes was to demonstrate that there is nothing unique about majority rule, the single alternative that is most often associated with nonunanimous collective action. But the problem of determining the rule to be chosen for postconstitutional collective choice is not my concern at this point. For present purposes, we may assume that any less-than-unanimity rule has been adopted, and we may use simple majority voting as illustrative. This rule is, presumably, chosen as one part of the more inclusive constitutional contract that defines the whole set of individuals' rights. My concern is the reconciliation of this majority rule with the multiparty exchange concept. What are individual "rights" in this setting? Can we discuss collective choice in terms analogous to voluntaristic exchanges

12. In the approach taken, there and here, unanimity offers the benchmark from which departures are dictated by reasons of efficiency in decision-making. Consent or agreement remains the conceptual ideal. For an argument which opposes this approach, see Douglas W. Rae, "The Limits of Consensual Decision" (Paper presented at the Public Choice Society Conference, College Park, Maryland, March 1973).

in private goods? Is it necessary to analyze collective decisions in a framework that is entirely different from that which is applicable for unanimity rule?

Under a unanimity rule, decisions if made at all are guaranteed to be efficient, at least in the anticipated sense. Individual agreement signals individual expectation that benefits exceed costs, evaluated in personal utility dimensions, which may or may not incorporate narrowly defined self-interest. With a purely public good, the individually secured benefits, as evaluated, must exceed the individually agreed-on share of costs, measured in foregone opportunities to secure private goods. From an initial imputation of endowments or goods, the multiparty exchange embodied in public-goods provision moves each individual to a final imputation, which includes public goods, that is evaluated more highly in utility terms. Each person in the collectivity moves to a higher position on his own utility surface, or thinks that he will do so, as a result of the public-goods decision reached by unanimous agreement.

No such results are guaranteed when collective decisions are made under less-than-unanimity rules. Under simple majority voting, for example, a person may find that a majority decision for public-goods provision shifts him to a lower rather than a higher position on his utility surface. What are his "rights" in such a postconstitutional change? It would seem that, for the person in question, this sort of change could hardly be called a "contract." Goods that he values are taken from him against his expressed desire. Coercion is apparently exercised upon him in the same way as that exerted by the thug who takes his wallet in Central Park. This manner of speaking is commonplace, but it tends to obscure much that requires careful analysis. The thief takes the victim's wallet. We should agree that genuine coercion is involved here because the victim, the thief, and external parties agree and accept the property rights. The wallet was the victim's by right of assigned and acknowledged ownership. Is this comparable to the situation of the citizen who finds that he must, on fear of punishment, pay taxes for public goods in excess of the amounts that he might voluntarily contribute? Is the collectivity, acting as directed by the effective decision-making coalition authorized in the conceptual constitutional contract, analogous to the thief? There is no question but that the collectivity is perceived in this image by many persons, and not

only by those whose utilities may be directly reduced at a particular point in time.

This is one of the major sources of confusion in modern discussion of social policy, and it is related to a paradox of government that we shall examine in more detail in Chapter 6. If, as we have postulated, individual rights are defined as rights to do things with respect to some initial set of endowments or goods, *along with* membership in a collectivity that is empowered to act by less-than-unanimity rules, and, further, *if* these rights should be mutually accepted, it becomes inconsistent and self-contradictory for a person to claim that his "rights" are violated in the mere working out of the collective decision rules that are constitutionally authorized. At this point it is worth recalling once again that the analysis remains timeless. We are assuming that the *same* persons participate in the conceptual constitutional contract and in postconstitutional adjustments. From this it follows that, if a constitutional contract is made that defines separate persons in terms of property rights, and if these rights are widely understood to include membership in a polity that is authorized to make collective decisions by less-than-unanimity rules, each person must have, at this prior stage, accepted the limitations on his own rights that this decision process might produce. (Note that this statement need not imply that the prior constitutional contract was itself optimal or efficient. Note further that the justice or injustice of this contract is irrelevant here.)

To clarify the analysis, it will be helpful to distinguish two institutional structures of departure from a unanimity rule for collective action. In the first, collective decisions are made by less than full agreement of all members of the community, but these rules are externally constrained so as to guarantee outcomes that might, conceptually, have been attained under unanimity, without bargaining or agreement difficulties. That is to say, outcomes generated by collective choices must dominate the prechoice positions for all members of the community, evaluated in a utility dimension. In this restricted framework, it seems legitimate to refer to collective action as indirect *contract* or *exchange*. The decision rule embodying less than full agreement is necessary to avoid the behavioral effects of a unanimity rule, but the intent of the substitute rule is to accomplish essentially similar purposes. Even if an individual might have chosen differently from that outcome which the sub-

stitute rule produces, he has made a net improvement in his utility through participation in the collectivity. As later discussion will suggest, this restricted departure from unanimity is not without real-world application.

In the second set of institutions to be examined, collective decision rules are unconstrained, and when unanimity is dropped an individual may find himself actually suffering net utility losses from “participating.” That is to say, he may end up at a lower utility level than he might have been able to sustain in the complete absence of collective action. (Recall that we are continuing to assume that there has been mutual acceptance of initially defined rights to endowments.) It would seem to be an improper use of language to call this process “contractual.” In this case, collective action may seem to an individual to be equivalent to that taken by the thug in the park, or worse. Even here, however, care must be taken to specify just what protections the constitutional contract offers the individual against exploitative collective or government decisions. If the constitution embodies unconstrained collective action under less-than-unanimity rules, the individual does not really “own” the initial endowments or stocks in a manner at all analogous to “ownership” under the contrasting institutional structure. “Private ownership” takes on a wholly different meaning in this setting, a meaning that must be explored in considerable detail. Before such exploration, however, it will be useful to present the alternative structure somewhat more systematically.¹³

INDIRECT CONTRACT UNDER LESS-THAN-UNANIMITY DECISION RULES

We may discuss the first of the two alternatives with the aid of a simple two-person model and a single diagram. In figure 3.1 we measure the utility of one person, A, on the ordinate, and the utility of the other person, B, on the abscissa. (The construction is similar to that of fig. 2.2, chap. 2 above.) The utility attained by each person from the establishment of constitutional contract is shown at C. Trade between the two persons in divisible or private goods shifts the position to E. There remain, however, further gains-from-

13. The whole discussion here is directly related to the issues involved in “just compensation.” These are examined in the context of modern legal doctrine in an excellent long essay by Frank I. Michelman. See his “Property, Utility, and Fairness,” pp. 1165–1258.

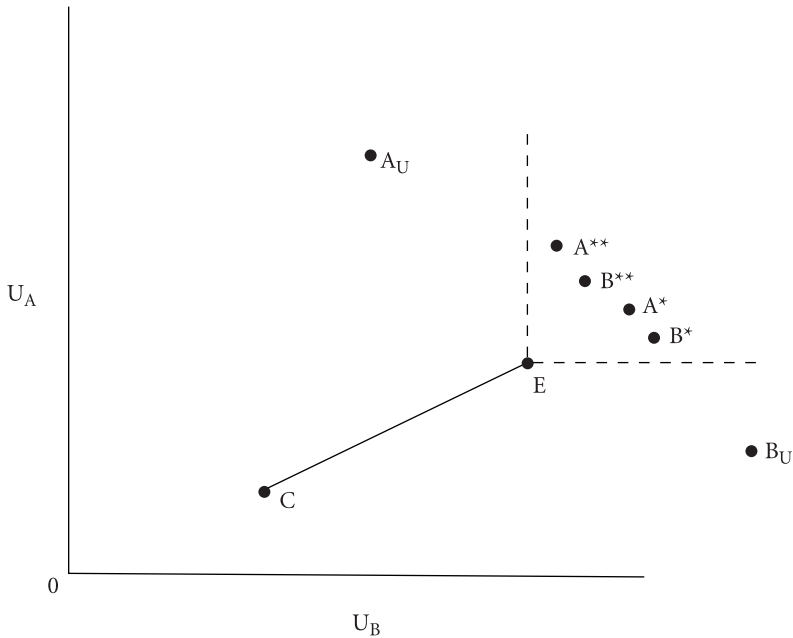


Figure 3.1

trade to be secured from the provision of a good that is consumed jointly. (The two-person model here is treated as being analogous to a many-person model; in an actual two-person interaction, few problems would arise in reaching agreement on joint or collective sharing.) If the group can attain agreement under a unanimity rule, this insures that a final outcome will be contained in the northeast quadrant from E, the area bounded by the dotted lines extending from E. In our extension to the many-person world, however, we have assumed that an effective unanimity rule is unworkable and that some less-than-unanimity variant is adopted constitutionally. In the two-person illustration, this means that decisions for the group will be made by either one or the other of the two persons, by either A or B, and independently of the preferences of the other.

We assume that collective decisions are limited to the purchase, provision, and financing of a single purely public good. Note that this assumption, in itself, severely constrains the set of outcomes that are possible. At best, the single ruler could impose all costs of the good on his cohort while providing

the good out to his own satiety levels. This constraint alone would not, however, normally be sufficient to insure that outcomes fall within the indirect exchange area bounded by the dotted lines of figure 3.1. As a further constraint, let us assume that the basic constitutional contract also specifies a taxing institution. That is to say, there exist constitutional requirements that the single public good must be financed from a specific tax structure. We might select any one of the familiar patterns of taxation such as equal-per-head charges, proportional taxes on incomes, progressive taxes on incomes, or others. In this setting, if the tax institution is properly chosen, the results generated under one-man rule may be beneficial, in the net, to both persons. Ideally, choice of the tax structure could make the collective decision rule irrelevant since all rules would produce the same outcome.¹⁴ The ideal tax could not, of course, be selected at the constitutional level. But with some practicable choice of tax structure along with the limitation of collective action to the provision of genuine public goods, we might plausibly predict that outcomes would be bounded within the Pareto-superior area. Consider, as a supplement to figure 3.1, the demand patterns shown in figure 3.2, and suppose that equal-per-head taxes are required. If individual A is the ruler, he will choose a quantity Q_a ; if individual B is the ruler, he will choose Q_b . Note that, in each case, the nonruler will still be enjoying a net fiscal surplus from participating in the public-goods arrangement. We might depict these separate outcomes in figure 3.1 as positions A^* and B^* respectively. Note that both fall well within the boundaries of the Pareto-superior set.

This means that either A^* or B^* could conceptually have been achieved from the working of a unanimity rule, given the chance pattern of bargaining toward solution that might have generated such outcomes. Given the tax institution postulated, the nonruler, B, will not be satisfied at A^* , or Q_a . He will not be in full marginal adjustment with respect to public-goods quantity and

14. Wicksell was the first to recognize the basic substitutability between tax institutions and collective-choice rules. By the introduction of more flexibility in tax institutions, more inclusive collective-decision rules can be accepted, with more guarantees against fiscal exploitation. Wicksell did not, however, reverse the logical chain here. With effectively designed tax institutions, the potential exploitation that can be implemented through less-than-unanimity decision rules can be reduced, and, in the limit, wholly eliminated. For an elaboration of this relationship, see my *Demand and Supply of Public Goods*, and my *Public Finance in Democratic Process* (Chapel Hill: University of North Carolina Press, 1967).

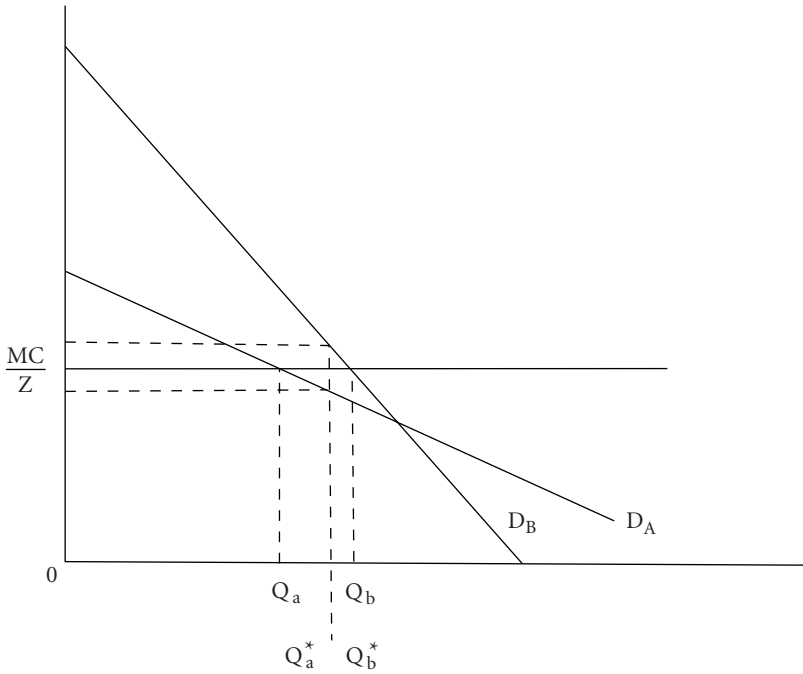


Figure 3.2

tax-price. If he should be made ruler, he would prefer to shift to B^* , and Q_b . He will, therefore, be “unhappy” with the tax-budgetary decision imposed on him by A through the solution at A^* . The same results would apply conversely to A if he should be the nonruler.

Differing constraints will, of course, generate different results, even under identical decision rules. Assume now that A remains the collective decision-maker, but that instead of head taxes, proportional income taxes are required. Furthermore, assume that B has a higher income than A, as indicated by the positions of D_A and D_B in figure 3.2. This tax will tend to reduce the quantity of good preferred by B and to increase that preferred by A. Under this scheme of taxation, individual A might optimally choose Q_a^* and individual B might choose Q_b^* . The utility positions attained under the alternative single-man rule are shown as A^{**} and B^{**} in figure 3.1. As depicted, both remain within the Pareto-superior region with relation to the initial position, E. This example suggests that there may exist a whole set of tax and budgetary insti-

tutions, or, more generally, constitutional constraints on fiscal process, which will insure that nonunanimity rules operate effectively as instruments to produce what we have called “indirect exchange” among individuals for purely public goods.

It is important to recognize both the purpose and the limits of the constitutional constraints that may be imposed on the operation of nonunanimity rules for collective decision-making at the postconstitutional stage of social interaction. To remain within what we may call broad contractual bounds, individuals must be assured that, in the net, operational politics will produce for them benefits rather than damages. There is nothing in the public-goods “exchange structure,” however, that dictates uniqueness in the distribution of the gains-from-trade, nothing analogous to the unique distribution of the comparable net realizable surplus in private-goods trade, no unique price vector emergent from idealized recontracting. For this reason, there may be considerable variation in political-institutional structure, in rules, without forcing results out of the bounds of mutuality of benefits among all parties. In game-theoretic parlance, the core of the public-goods game is considerably more inclusive than that in the private-goods game.¹⁵

In a regime with perfect side payments and zero transactions costs, a unique allocative result for the provision of a purely public good will be generated only if income-effect feedbacks are neglected or absent. However, any allocative result achieved may, itself, be attained from any one of a whole set of distributive patterns. An allocative result requires that marginal prices confronted by separate participants stand in some specific relation one to another. There is no comparable relationship among average prices. The relatively loose restrictions imposed by the “indirect exchange” constraints require only that all persons secure net benefits. In the more general setting where

15. Formal proofs that the Lindahl “solution” to the public-goods game is in the core do nothing toward showing that there do not exist many other “solutions” that equally qualify for inclusion. In this sense, the Lindahl equilibrium in the public-goods game is not at all comparable to competitive equilibrium in the private-goods counterpart.

To make the geometrical illustration in figure 3.1 more relevant to the large-number setting, we could reinterpret position E as that which is attained from the completion of all interpersonal trades in private goods and in all trades in joint-consumption goods that involve coalitions of less than the full membership of the inclusive group. With this modification, the diagram and the discussion can depict the constraints imposed on political decision rules with respect to any one person vis-à-vis all others in the polity.

side payments are costly, the constraints also allow for considerable departures from the attainment of idealized allocative efficiency. The necessary condition is only that public-goods exchange, conceived as a game, be positive sum for all participants. There is no necessity that aggregate payoffs be maximized. To the extent that total payoffs may be influenced by the rules, one criterion of adjustment at the constitutional stage becomes that of predicted allocative efficiency. But, as we shall suggest, this criterion may well be dominated by distributive norms.

UNCONSTRAINED DEPARTURES FROM UNANIMITY RULES

A categorically different model is introduced when collective decisions can be taken under nonunanimity rules with no constitutional limits or constraints. Recall that our basic schemata includes a conceptual separation between the stage of constitutional contract, at which individual rights are defined and collective decision rules are made, and postconstitutional contract, at which trades or exchanges take place among persons whose rights to carry out activities and to dispose of things are defined in the prior stage. This schemata allows us to discuss market process, private-goods trade, and those political processes that embody public-goods “exchanges” in the postconstitutional stage. We have incorporated the “indirect exchanges” that take place under constrained nonunanimity rules for collective choice. If, however, the delineation of rights in the constitutional stage allows the collectivity, the state, to make decisions on any less-than-unanimity rule without *constraint*, the proposed schemata appears to involve an internal contradiction. In the two-person example, we could scarcely argue that B’s rights are defined in some prior stage if A’s rights are all-inclusive and unrestricted. The model emphasizes the necessity of defining the “rights” or limits of the decision-maker for the collectivity as well as those of the separate persons within it. So long as the collectivity’s actions are constrained as noted above, we are able to talk as if the contracting parties in postconstitutional negotiations are individual citizens. Although vastly more complex, political process becomes analogous to market process. But we are no longer able to think in such terms when all shackles are removed from collective action.

An attempt might be made to get out of the apparent contradiction here

by resort to “probabilistic rights.”¹⁶ That is to say, we might consider that human and nonhuman rights are defined in the constitution subject to the actions to be taken by an unconstrained collectivity, operating under any of a large subset of specified nonunanimity rules for decision, ranging from the near-unanimity of a Wicksellian qualified majority, through simple majority voting, to one-man dictatorship. The value of any individual’s actual claim over “goods,” including over life itself, might then be represented by some “expected value,” determined by the descriptive characteristics of the decision rule in being, by the social history of the collectivity, by the value of the nominal claim, and by the probability that this claim will be changed, upward or downward, by imposed action taken in the name of state or collective authority.

Consider, for example, the position of a person who, in nominal terms, has been assigned command over a relatively large share of “goods” in the community, but who holds membership in a collectivity that makes decisions on the basis of simple majority voting, without explicit or traditional constitutional constraints. In this setting, the probability of the person in question being able to retain the full nominal value of his “goods,” as assigned, or to improve this by participating in public-goods trade, might be low. He should find it possible to compute some plausibly realistic “expected value” for his nominal claims. The question is whether this value might offer a basis for both private- and public-goods trades or exchanges. To the extent that trades can take place in terms of such expected values, it seems evident that risk elements will necessarily dampen pressures toward efficiency. A more fundamental problem arises, however, concerning the nominally assigned claims upon which such expected values might be computed. If the contractual setting is literally applied, why should a person have accepted the unconstrained collectivity in juxtaposition with his relatively favorable set of nominal rights? Even conceptually, why should he have ever acquiesced in the assignment of unlimited rights to the collectivity? These issues force us back into a discussion of constitutional contract itself which we have tried to relegate to a subsequent chapter. But a provisional answer may be advanced

16. For a related discussion in the context of theoretical welfare economics, see A. Mitchell Polinsky, “Probabilistic Compensation Criteria,” *Quarterly Journal of Economics* 86 (August 1972): 407–25.

here. If an individual acknowledges the existence of an unconstrained collectivity that reduces the expected value of his net claims, he should rationally have preferred some initially defined constitutional reduction in his nominal claims along with an accompanying restriction on collective action. Similarly, if another person finds that the expected value of his net claims, under the operation of unconstrained collective action, exceeds that measured by the nominal value of his assignment, he should prefer a somewhat larger nominal assignment along with imposed limits on the collectivity. For both persons, uncertainty is reduced by restrictions on state action.

This would provide a logical basis for the imposition of constraints on the collectivity, as an acting unit, even if institutional necessity requires that this unit act independently from individual efforts. An additional basis for constitutional constraints on collective action is provided when it is recognized that, if there are no constraints, individuals have a stronger incentive to invest resources in attempts to secure control over collective decisions. Control over the collective decision-making apparatus becomes the instrument for securing the winnings of a zero-sum component of the game of politics. And, for the community in total, all resources invested in gaining this control are wasted. Incentives to gain control over the collective decision-making machinery are not, of course, wholly absent in the fully constrained model. As the simple diagram in figure 3.1 shows, it does make some difference in utility terms whether A or B is the effective decision-taker for the community. In the unconstrained model, however, individual A might look on the prospects of reaching some position like A_u that would result from his gaining control over the decision-making for the collectivity, while B might be similarly attracted by the prospects of moving to B_u if he successfully seizes the reins of government. It is clear that the incentives to invest resources in "politics" become dimensionally larger in the unconstrained than in the constrained model.

Corollary to this is the motivation for persons who control collective decision-making to use this means of generating directly enjoyable and divisible private and partitionable goods rather than producing genuine public goods which benefit all persons in the community. In a constitutionally unconstrained collectivity, it seems likely that net wealth and income transfers would bulk much larger in governmental action than they would in constrained constitutional regimes. The use of tax revenues collected from those

who are the “outs” to finance Swiss bank accounts for the “ins” is the familiar real-world example.

In a sense, the analysis of unconstrained collective action under nonunanimity decision-making brings us full circle. The very purpose, in the larger “social” meaning, of defining rights in constitutional contract is to facilitate orderly anarchy, to provide the basis upon which individuals can initiate and implement trades and exchanges both of simple and complex forms. Having defined and accepted a structure of rights, individuals can reduce their own investment in defense and predation and go about their business of increasing utility levels through freely negotiated dealings with each other. To the extent that collective action is allowed to break beyond the boundaries imposed by the mutuality of gains from exchange, both direct and indirect, the community has taken a major step backward into the anarchistic jungle or has failed to take the major step from this jungle in the first place.

The operation of an unconstrained collectivity could scarcely emerge from rational constitutional contracting among persons. Historically, an explicit stage of constitutional contracting may never have existed; the structure of rights may have emerged in an evolutionary process characterized by an absence of conscious agreement. From this setting, the apparent contradiction may be generated. More important for my purposes, even if something akin to an initial contract may have settled the structure of individual and collective rights, this structure may be eroded over time. Although once constrained, the powers of the collectivity may gradually be expanded so as to become, for all practical purposes, unlimited. As we have noted earlier, the contractual models are not designed to be historically descriptive. They are, instead, designed to assist in the development of criteria with which existing political-legal systems may be evaluated. In this context, empirical evidence that the collectivity as it exists is unconstrained suggests the hypothesis that general agreement could be attained for genuine constitutional revision.

The analysis in this chapter, and elsewhere, is derived from the basic norms of individualism discussed in Chapter 1. The position taken here stands in apparent opposition to the allegedly “positivist” view that denies the possibility of constraining the collectivity in any ultimate sense. This was Hobbes’s position, and in his conceptual model the individual surrenders all rights to the sovereign at the time of the initial contract. In the terminology employed here, this amounts to saying that only the collectivity, the government, holds

anything that might be called “rights.” Those claims to carry out specific activities, including disposition and usage of resources, that are expressed on the part of persons are subject at all times to arbitrary redefinition by government. And, indeed, the central role of government in this positivist model is the settlement of competing claims among individuals and groups, settlement which necessarily involves continual redefinition of limits.¹⁷ I shall not defend the approach taken in this book against the positivist arguments. Whether it is possible to constrain the powers of government, to protect individual rights in a genuine usage of this term, can never be proven empirically. It is at this point, however, that individuals’ attitudes toward reality seem more important than the reality itself. Governmental decisions are always made by men, and if these men act within a paradigm that embodies meaningful constitutional constraints, the “as if” analysis seems warranted, regardless of the ultimate power that may or may not remain unexercised.

Allocation and Distribution

The categorical distinction that I have made between constitutional contract and postconstitutional contract may seem familiar to economists. The distinction is related to the familiar neoclassical dichotomy between allocation and distribution, especially as the latter is treated in normative discourse in political economy. In the world restricted to private or partitionable goods and services, when property rights are defined, markets will emerge to allocate resources with tolerable efficiency and the gains-from-trade will be distributed among particular parties in a specific manner. Neoclassical, and mod-

17. Warren J. Samuels is an articulate modern spokesman for this positivist position. In an exchange devoted to a specific legal issue, the basic methodological differences between this position and my own are clarified. See Warren J. Samuels, “Interrelations between Legal and Economic Processes,” *Journal of Law and Economics* 14 (October 1971): 435–50; James M. Buchanan, “Politics, Property and the Law: An Alternative Interpretation of *Miller et al. v. Schoene*,” *Journal of Law and Economics* 15 (October 1972): 439–52; Warren J. Samuels, “In Defense of a Positive Approach to Government as an Economic Variable,” *Journal of Law and Economics* 15 (October 1972): 453–60.

A more comprehensive statement of Samuels’s position is contained in his “Welfare Economics, Power, and Property,” in *Perspectives of Property*, ed. G. Wunderlich and W. L. Gibson, Jr. (State College: Institute for Land and Water Resources, Pennsylvania State University, 1972), pp. 61–146.

ern, economists have expressed little or no direct concern for the market's distribution of the *gains*-from-trade. They have been unwilling to accept the final distributional results largely because they remain unwilling to restrict their domain of evaluation to postconstitutional contract. The distinction developed here would have been helpful in clarifying much of the discussion in political economy because this would have indicated that the distributional problem arises, not with respect to the gross gains-from-trade, but with respect to the initial distribution of endowments or capacities—that distribution that provides the basis upon which individuals enter the trading process.

In this particular respect, discussion and analysis of public-goods exchange in postconstitutional contract have been considerably more sophisticated than the parallel analysis of the private market sector. Knut Wicksell recognized explicitly that the efficiency norms for the provision of jointly consumed goods and services which informed his search for the appropriate institutions for collective decision-making are applicable only in a setting where individual property rights are well defined and broadly acceptable. Wicksell recognized that the whole decision-making process must be modified when genuine constitutional contract is considered. In this respect, as in others, my own approach has been greatly influenced by Wicksell. In modern public-finance theory, R. A. Musgrave, in his basic treatise, makes a categorical distinction between the allocational branch of the budget and the distributional branch.¹⁸ Especially in his response to attempts to extend allocational norms to distributional policy, Musgrave seems to make a categorical distinction between the fundamental decision processes that are involved.¹⁹

18. R. A. Musgrave, *The Theory of Public Finance* (New York: McGraw-Hill, 1959).

19. R. A. Musgrave, "Comment," *American Economic Review* 60 (December 1970): 991–93. The paper that prompted Musgrave's comments, along with those of several others, was Harold M. Hochman and James D. Rodgers, "Pareto Optimal Redistribution," *American Economic Review* 59 (September 1969): 542–57.

4. Constitutional Contract

The Theory of Law

As suggested above, postconstitutional contract has occupied the primary attention of economists for the whole period of their independent disciplinary existence. Despite the concentration of effort on the exchange processes, major analytical complexities remain unresolved. What then may we anticipate when we try to conceptualize constitutional contract, that human interaction in which individual rights may be initially defined, where the very rules for interpersonal behavior may be established, where “society,” quite literally, replaces “anarchy”? Once we so much as open up this area for critical examination, is there much surprise in professional economists’ proclivity to commence with the assertion that property rights are well defined? Is there any wonder that a genuine economic theory of law remains undeveloped?¹

Strictures should not, however, single out economists. They can with some legitimacy claim exemption; their traditional domain is or should be limited to contract. Perhaps criticism is more appropriately directed at those whose professional emphases are on power relationships among individuals and groups. But political scientists have been reluctant to follow up the leads suggested by Thomas Hobbes. They have devoted much attention to the politi-

1. A development of significance in the 1960s and 1970s has been the emergence of the “economics of property” as a field of intense scholarly interest, along with the parallel introduction of more economic theory and more economists into the curricula of law schools. The primary emphasis of this movement has been, however, the influence on individual behavior exerted by the institutional or legal setting within which such behavior takes place. This emphasis is, in itself, praiseworthy, and it has yielded and will continue to yield useful scientific results. By contrast with this, however, relatively little emphasis has been placed on the possible explanations for the emergence of the observed institutions in the first place. See relevant footnotes to Chapter 2.

cal obligation of individuals, leaders and followers alike, but relatively little attention to the base positions from which behavioral obligations must be assessed. In its own defense, however, political science can claim to embody a more developed and sophisticated historical sense than economics. Once it is recognized that observed institutions of legal-political order exist only in a historical setting, the attraction of trying to analyze conceptual origins independent of historical process is severely weakened. The temptation becomes strong to assert what is essentially the positivist position that a structure of law, a legal system, a set of property rights, exists and that there is relatively little point in trying to understand or to develop a contractual metaphor for its emergence that would offer assistance in finding criteria for social change. There is merit in this approach, provided that it is not allowed to exclude complementary bodies of analysis. Some of the implications of accepting "law as fact" will be explored in Chapter 5.

In order to discuss or to analyze possible criteria for modifying the structure of rights, however, some understanding of conceptual origins may be helpful. As has been suggested, the problem is one of trying to explain and to understand the relationships among individuals, and between individuals and the government. And for this purpose, various "as if" models of conceptual origins may be necessary, regardless of the facts described in the historical records.² Stress should be placed on "explanation" and "understanding" since the temptation to introduce normative statement becomes extremely strong at this level of discourse. Precisely because the conceptual origins are discussed independent of observable historical data, the distinction between positive analysis and normative presupposition is difficult to detect.

Must we postulate a basic equality among men in some original setting in order to derive the structure of a free society from rational, self-interested

2. See S. I. Benn and R. S. Peters, *The Principles of Political Thought* (New York: The Free Press, 1959), p. 377.

Maine advanced the suggestion that the language and terminology appropriate to discuss the relationship between ruler and ruled in postfeudal society was found only in treatments of Roman law. Hence, "contract" theories of the state emerged, in part, because of linguistic history. In this connection, Maine's discussion of "quasi-contracts" is helpful, and this would perhaps be a more suitable term to apply to all discussion of "social contract." In "quasi-contract," there is no implication of explicit agreement, but the relationship is such as to make the contractarian framework for discourse helpful. See Henry Maine, *Ancient Law*, pp. 333–35.

behavior? We have often answered this question affirmatively, even if implicitly. In the process, we have made our whole “theory” of conceptual constitutional foundations highly vulnerable to positivist refutation. In this book, I am attempting to explain how “law,” “the rights of property,” “rules for behavior,” might emerge from the nonidealistic, self-interested behavior of men, without any presumption of equality in some original position—equality either actually or expectationally.³ In this effort, I make no claim to have escaped all normative influences. But I should argue that the approach taken is less normative than the familiar one which says, in effect, that any logical analysis of law *should* be based on the *as if* presumption of personal equality. We can substantially strengthen the foundations of freedom if we can succeed in demonstrating that, even among men who are unequal, a structure of legal rights can be predicted to emerge, a structure that retains characteristic elements that we associate with the precepts of individualism. Only after we have done this can we begin to offer constructive criticism of the veritable maze of confusion about constitutional order that abounds at the most basic levels of discussion.

Personal Inequality

In order to make an analysis of constitutional contract as general as is possible, allowance should be made for the existence of substantial differences among persons in the original conceptual setting. This is not the same as postulating inequality as fact. The analysis should be sufficiently general to be applicable if, in fact, persons should prove to be substantially equivalent. To be avoided is the dependence of the results on some unsupported presumption of natural equality. The degree or measure of inequality will, of course, affect the description of any initial position and the structure of rights that may contractually emerge.

Consider, then, some initial setting in which men are not equals. Following economists’ practice, we can discuss inequality in two separate attributes:

3. It is in the expectational sense that the approach taken here differs sharply from that taken in *The Calculus of Consent*. In that book, individuals were assumed to be sufficiently uncertain about their positions under the operations of decision rules as to make them enter the negotiation stage as equals in at least this one respect.

(1) tastes or preferences, and (2) capacities.⁴ It is necessary to avoid explicitly the tendency to slip exclusively into the familiar classification of persons by personal endowments of “goods,” presumably measured in commodity dimensions. This procedure amounts to neglecting the very problems addressed in this chapter, to presuming that individual rights to commodities, to goods, have already been defined.

In the fundamental sense required for the analysis here, an individual possesses no “goods” or “resources.” He can be defined initially by a preference or a utility function on the one hand and by a production function on the other.⁵ The preference or utility function describes the rates at which the person is willing subjectively to trade off goods (and bads) one against the other. The individual’s production function is less familiar. He will have, inherent in his physiological makeup, a set of capacities (skills, talents, abilities). These capacities, when exercised in a specified environmental setting, define for the individual a potential relationship between inputs (negative goods or bads) and product (positive goods). This relationship is his “production function.”

As noted, persons may differ from each other either in tastes, in capacities, or in both. Or, persons identical in both tastes and capacities may find themselves in environmentally different situations relative to their capacities. A person with mediocre talents may be confronted with ample opportunities to secure positively valued goods while a person with superior talents may face less favorable opportunities. The position attained by a person is dependent on three basic elements: his preferences, his capacities, and his environmental setting. It would be wholly arbitrary to assume that all individuals face identical environmental settings; this would be as indefensible an assumption as that which postulates personal equality in preferences or in capacities.

Anarchistic Interaction

Consider two individuals who are wholly isolated from each other; each on his separate island with no social contact. Each man would attain a personal

4. For an analysis of the different implications of inequality in these separate attributes, see my paper, “Equality as Fact and Norm,” *Ethics* 81 (April 1971): 228–40.

5. Note that we are removing the restrictions imposed by the no-production model analyzed briefly in Chapter 1.

behavior equilibrium, as determined by the interaction among his utility function, his basic or inherent capacities to convert input into output, and the natural environmental setting that he confronts. There would be no easy means of judging which of these two Robinson Crusoes is more favorably situated, or which one secures more "welfare." This two-Crusoe world is, of course, purely anarchistic. There is no law, and there is no need for a definition of individuals' rights, either property rights or human rights. There is no society as such. Nonetheless, this two-Crusoe world provides a useful starting point from which to begin consideration of the world where personal conflict may emerge. Suppose that the persons, whom we shall name A and B, no longer exist in complete isolation, but that, instead, they now find themselves in some spatially limited area, on the same island. This change, in itself, need not modify the preferences of either person, although such an effect should not be ruled out. The environmental setting of each person will, however, almost surely be modified. In the absence of law, each person will now consider the other as a part of the environment that he faces. The effects of this upon the rate at which bads may be transformed into goods may take on several patterns.

In a world of scarcity, mutual exploitation of the natural environment insures that, for each person, the terms-of-trade with his own environment are worsened relative to those confronted in the isolated setting where one person, alone, faces this environment. In effect, the natural environment becomes "common property," and the familiar reciprocal externality relationships emerge. Most economists would perhaps tend to stop the analysis at this point with little or no consideration of the remaining possibilities. But a second, and very different, sort of influence may operate. If production is not simultaneous with actual consumption of goods, individuals may store goods for future use. In this situation, the presence of B may prompt A to devote effort, a bad, to concealing hoards, and to defending and protecting these hoards from predation by B. Since this effort might otherwise have been used to produce goods directly, A's net rate of transformation is adversely affected by this necessity for defense.

An offsetting effect may, however, work in the other direction. Because of the presence of B, A now has available to himself a new opportunity. He may secure goods that were not available to him in the strict Crusoe setting. If B is known to be producing, and storing, goods, A may find that locating and

taking these stocks from B is more productive than producing similar goods on his own. This effect, if it should predominate, tends to shift A's production function in a favorable manner. Once the prospects for defense and predation are recognized, it is clear that individuals may differ in their talents for these activities and that such differences need not be directly correspondent to their relative capacities as direct producers. Furthermore, individuals may differ in their tastes for defense-predation efforts relative to direct production efforts.

It is, of course, impossible to consider the effects of B's presence on A without, at the same time, considering the effects of A's presence on B. The two persons are necessarily in a reciprocal interaction; their behavior is interdependent even if there is no social structure within which interdependence takes place. As indicated in Chapter 2, this sort of interaction may be analyzed in externality terms, even if we are working with a model without law and rights of property. It is useful to think of the reciprocal external diseconomy model in which each person's behavior imposes harm on the other. Consider, first, the behavior of A in producing goods in the environment that is shared, but without overt dispute. That is to say, let us postulate initially that A and B allow each other to attain private adjustment independent of disturbance. Each man uses his talents as best he can to maximize utility on the assumption that the other will not take stocks from him and that he, in turn, will not take stocks from the other. This is strictly an arbitrary starting point, and it will not represent a final equilibrium in the interaction sequence. In figure 4.1, this no-conflict position is placed at the origin. Note that, at this position, A and B need not have equal quantities of the good, nor need they be accepting equal quantities of the bad to secure the position indicated. For purposes of illustration, let us assert that at the origin, A is exerting six units of effort (a bad) to get, in the net, ten units of bananas (a good), while B is exerting five units of effort in getting twelve units of good.

This arbitrary starting point does not qualify for a behavioral equilibrium because, in this position, each person has some incentive to initiate conflict, to engage in predatory activity vis-à-vis his cohort. Figure 4.1 shows that, if A thinks that B will remain in the position at the origin, he will initiate predatory action to shift toward position Y. Conversely, B will be motivated to try to reach position X. The reactions of A to every level of B's activity in defense-predation is shown by the ridge line, R_A . Similarly, B's reaction to every level

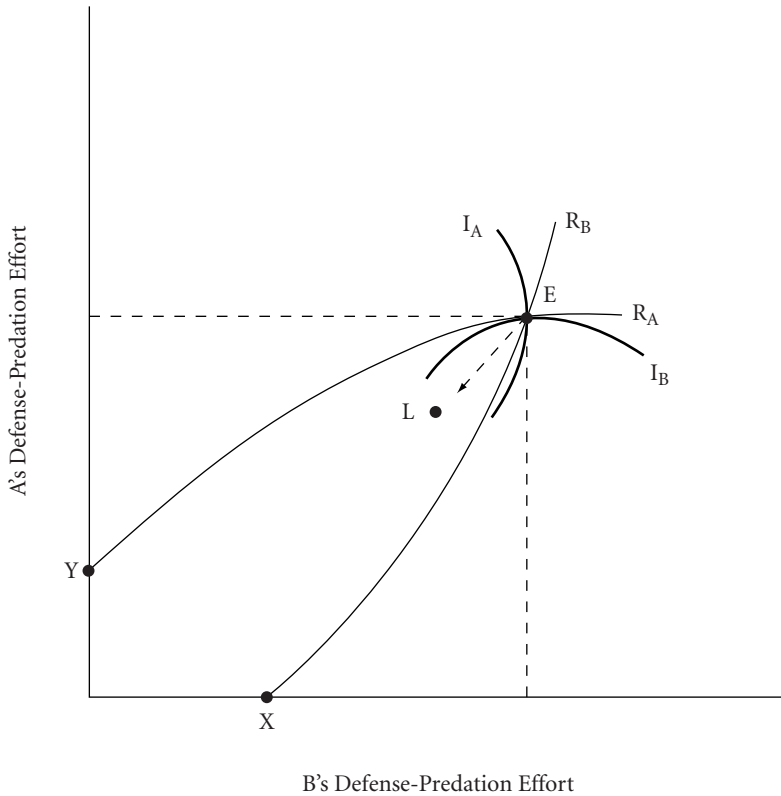


Figure 4.1

of defense-predation by A is shown by the ridge line, R_B . Equilibrium in this purely anarchistic setting is attained at E. At this point, neither person has an incentive to modify his behavior privately or independently. In this equilibrium, each person may be expending some share of his efforts in defending his stocks from the other, another part in taking stocks of the other, and another part in producing goods directly. The position of independent adjustment equilibrium describes the outcome or result that could be predicted in a genuinely anarchistic order. The distribution has been called the “natural distribution” by Winston Bush, and the two-person model may, of course, be extended to apply in a many-person setting.⁶

6. See Winston Bush, “Income Distribution in Anarchy.” See also Winston Bush and Lawrence Mayer, “Some Implications of Anarchy for the Distribution of Property,” mim-

I have emphasized in several places that there is no presumption of equality among persons in this independent adjustment or natural equilibrium. A second important principle is that this position cannot itself be attained contractually. Until this natural equilibrium is itself attained, there is no basis from which persons can negotiate contracts, one with another. The generation of this independent adjustment equilibrium is, therefore, the precontractual stage of social order, if indeed we can use the word “social” at all here.

There are no property rights in the strict sense of this term in the equilibrium so described. The position does, nonetheless, have certain stability characteristics, both for “society” and for the individual participant. There is no incentive for any person to modify his own behavior in the absence of exogenous shocks. In this equilibrium, therefore, each person knows with some degree of certainty what his own final command over ultimate consumable goods will be. Each person will, as noted, be expending resources in defending acquired stocks and in securing stocks initially acquired by others. But his net asset position, his final command over goods, will be predictable within relatively narrow limits. Chaos does not seem the appropriate descriptive word to apply to this genuinely anarchistic equilibrium if its meaning is taken to include unpredictability. Something akin to “property,” therefore, emerges from the noncontractual struggle in anarchy. Individuals achieve identifiable bases from which it becomes possible to make contracts.

Disarmament and the Emergence of Property Rights

In natural equilibrium, each person uses resources to defend against and to attack other persons. Each person would be better off if some of these resources could somehow be turned to the direct production of goods. The most basic contractual agreement among persons should, therefore, be the mutual acceptance of some disarmament. The mutual gains should be apparent to all parties.⁷

eographed (Center for Study of Public Choice, Virginia Polytechnic Institute and State University, Blacksburg, Virginia, 1973).

7. This derivation of the conceptual origins of property has been advanced by several social philosophers. It was developed by Aegidius Romanus in the thirteenth century, and

This may be illustrated with reference to the interaction depicted in figure 4.1. By the definition of ridge lines, or lines of optima, we know that the indifference contours for A are vertical along R_A , while those for B are horizontal along R_B . Hence, at E, we know that the indifference contours intersect at right angles, in the manner shown by I_A and I_B . We know, further, that the indifference contours for A are concave to the left, while those for B are concave downward. This is because of the fact that the ideal position for A is at Y, where B makes no effort to defend stocks or to take stocks from A, and that the ideal position for B is the comparable one shown at X. In these configurations, the Pareto-superior region, that which includes positions reflecting mutual gains by comparison with E, lies to the southwest, indicated directionally by the arrow in figure 4.1. Positions that embody mutual gains must involve lower outlay on defense-predation for both parties. Suppose that an agreement is reached to move to position L. Note precisely what this agreement embodies. The contract is one of bilateral behavioral exchange. Individual A agrees to give up some share of his own defense-predation effort in exchange for a related behavioral change on the part of individual B. There is no incentive for either person to take this behavioral change unilaterally, and there is nothing in the initial agreement, as such, which requires or even induces any acceptance by the other of the legitimacy of either person's command over goods, either in the preagreement or postagreement stage. Mutual acceptance of "ownership rights" is not a part of this preliminary disarmament agreement. On the other hand, by negotiating such an initial agreement to limit defense and predation, "law" of a sort has now emerged. The two persons accept limits to their own freedom of action, to their own liberty. The first leap out of the anarchistic jungle has been taken.

Conquest, Slavery, and Contract

In the discussion of anarchistic interaction to this point, I have assumed implicitly that all persons will exist as independently acting defenders and pred-

was elaborated with surprising sophistication by Hugo Grotius in 1625. On these contributions, see Schlatter, *Private Property*, pp. 57f., 128–32.

The derivation is also closely similar to that presented by David Hume. See his *A Treatise of Human Nature*, vol. 3.

The contractual emergence of property rights from some anarchistic state of nature is opposed to the view taken by many scholars that "natural law" precepts are necessary.

ators both prior to and after a natural equilibrium is attained. If personal differences are sufficiently great, however, some persons may have the capacities to eliminate others of the species. In this instance, the natural equilibrium may be reached only when the survivors exercise exclusive environmental domain.

The complete elimination of other persons may not, however, be the most preferred course of action by those who possess superior capacities. Even more desired might be the state in which those who are “weak” are allowed to exert effort in producing goods, after which the “strong” seize all, or substantially all, of these for their own use. From this setting, the disarmament contract that may be negotiated may be something similar to the slave contract, in which the “weak” agree to produce goods for the “strong” in exchange for being allowed to retain something over and above bare subsistence, which they may be unable to secure in the anarchistic setting.⁸ A contract of slavery would, as other contracts, define individual rights, and, to the extent that this assignment is mutually accepted, mutual gains may be secured from the consequent reduction in defense and predation effort. This may seem to represent a somewhat tortuous interpretation of slavery as an institution, but it is explicitly designed to allow the analytical framework developed here to be fully general.⁹

Trading Equilibrium and Direct Production

Economists who are familiar with the geometrical construction of figure 4.1, and the underlying postulates, will recognize that the minimal restrictions imposed on the locations and shapes of the indifference contours do nothing toward insuring that a final post-trade position where all gains are exhausted will be coincident with the origin, which describes the allocation or result that would obtain in the absence of all defense and predation effort. This direct-production position, in which each person retains for his own use

8. In terms of the model of figure 4.1, the natural equilibrium for one of the two persons would clearly be less desirable than the position at the origin, where, by assumption, he is allowed to retain what he produces.

9. For a different and more general discussion of the emergence of slavery, see Mancur Olson, “Some Historic Variations in Property Institutions,” typescript (University of Maryland, 1967).

those goods that he himself produces, given his own capacities, his tastes, and his environmental situation, may or may not be Pareto-superior to the natural equilibrium at E; and even if the direct-production position should qualify as Pareto-superior it need not lie along the contract locus which would be generated by trading among the two parties.

The relationship between E, the position of equilibrium attained in the absence of law, and the origin, that position attained when each man keeps all that he produces, is important because of the dominant role that has been assigned to the latter in the historical discussions of property rights, notably those discussions in the natural-law tradition, and especially as represented in the theory of John Locke. In the conceptual origins of contract that have been developed here, there is no fundamental distinction between the position which allows persons to retain goods privately produced and any other position. The only distinguishable position, prior to contract, is that shown in the natural equilibrium at E.¹⁰

If the direct-production position is Pareto-superior to E, by which we mean only that both parties secure higher utility levels in the former position than in the latter, there may well be a strong attraction toward settling the negotiations at this point, even if the direct-production position does not qualify as falling along the strict contract locus. There are two related reasons for this. In the first place, initial agreements on limiting behavior would not be likely to take place in terms of finely tuned marginal adjustments. Instead, a once-and-for-all quantum leap might be suggested, without the tedious bargaining required for sophisticated adjustment. In this sense, any position within the lozenge confined by the indifference contours would qualify for a settlement prospect. Among this large set of Pareto-superior positions, those which seem the most likely candidates for agreement will possess Schelling-point characteristics. Positions which qualify here are those that are simple and known to all parties and which will tend to be selected in the absence of information and communication between the interacting parties.¹¹ An agreement to eliminate all predatory behavior might be a plausible outcome un-

10. If there is something inherent in the nature of man that inhibits theft, the natural equilibrium and the origin on figure 4.1 would coincide.

11. See Thomas C. Schelling, *The Strategy of Conflict* (Cambridge: Harvard University Press, 1960).

der this setting, in which case each person's production from the natural environment that he confronts becomes his "property" in some positive sense. The law might begin to take on positive features in a manner akin to that rationalized by John Locke.

The predominant role that has been assigned to the direct-production position may be based on the implicit assumption of natural equality among men. If we allow interpersonal differences to exist in the natural state, however, there is no assurance that the position attained in the anarchistic equilibrium, depicted at E in figure 4.1, is Pareto-inferior to the direct-production position at the origin. The latter position need not lie within the lozenge enclosed by the indifference contours drawn through E. At least one of the two persons may be better off, in utility terms, in anarchistic equilibrium than he would be if required to depend exclusively on his own production efforts (as in the slavery example noted). This outcome might emerge if the two persons were widely different in the ability to produce goods, either from a difference in natural capacities or from a difference in environmental situations. Also, such an outcome might arise if one person retains moral inhibitions against predation while the other does not, or even if one person values liberty of action so highly that he willingly sacrifices protection of goods produced.

When the direct-production position is not Pareto-superior to E, positive property rights to goods directly produced will not emerge from conceptual contractual agreement. Something other than an agreement on mutual limits to behavior is required to leap from the Hobbesian jungle in this case. Such an agreement on limits must be accompanied by a transfer of goods or endowments before a contractual settlement can be reached, and property rights positively established.

This may be illustrated with a different geometric construction, although still within the confines of a two-party model. In figure 4.2, effort is measured along the ordinate, and goods along the abscissa. Individual A is either favorably situated or is more capable of producing goods than individual B. The production function for A, if he is not interfered with by B, is shown by the curve P_a , which lies along the abscissa for an initial range, indicating that A can secure some goods without an outlay of effort. Individual B, by contrast, faces a much more unfavorable direct-production prospect. In the absence of all interference from A, he faces the production function shown by

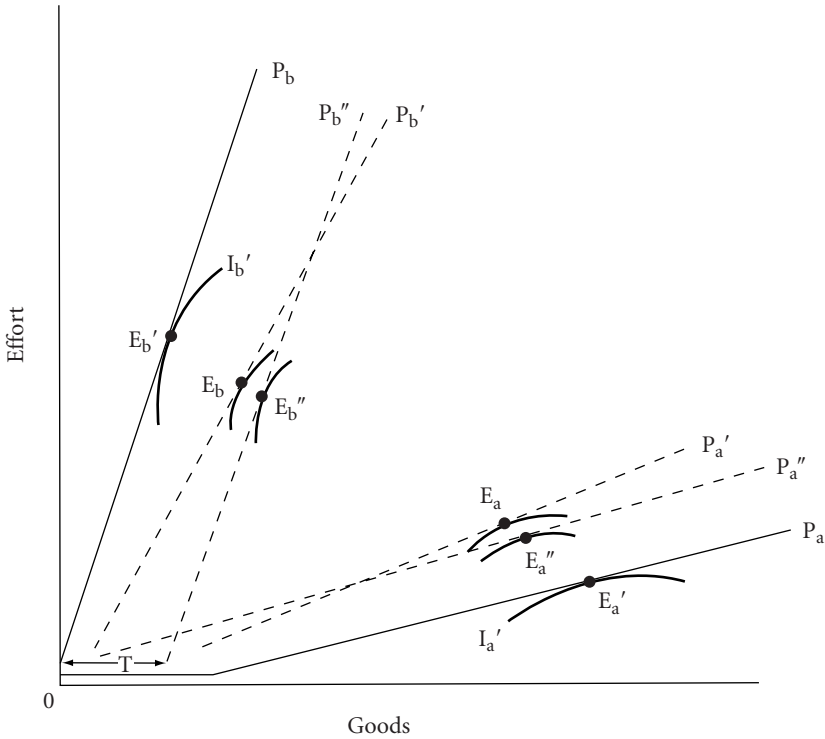


Figure 4.2

P_b . The direct-production position, that represented by the origin of the earlier figure 4.1, is attained when A attains point E_a' and when B attains point E_b' . In the situation where no rights of property are assigned, B may well find that his most productive expenditure of effort lies in predation, in stealing goods that are produced by A. If A undertakes no defense or protection effort, the anarchistic production function faced by B might be like that depicted by P_b' , along which B would move to position E_b . This activity on the part of B would, of course, modify the situation faced by A. He would, privately, face the production function shown by P_a' , if he undertakes no responsive action. In order to illustrate the relevant relationships in a diagram like figure 4.2, we shall assume that A does not find it advantageous to respond to B's predation. A's new equilibrium position would be that shown at E_a . Since we have assumed that A undertakes no defense or protection ef-

forts, his actual production function is not modified, but he is producing a portion of his goods for B. The anarchistic equilibrium is that position indicated by the two points E_b and E_a in figure 4.2. It is clear that, for B, this is a more favorable situation than that which he attains when property rights are assigned in goods that are directly produced. Hence, B would never agree to the direct-production position. Contractual arrangements must include something over and beyond limits to behavior. In this setting, A might achieve B's agreement to respect an assignment of rights to goods that are produced privately or independently if he transfers to B some initial quantity of goods or endowments. One such transfer can be depicted on figure 4.2 by the amount T, as indicated. If this is transferred to B, his direct production function shifts to P_b'' , and his attainable private production equilibrium to E_b'' , which is on a higher utility level than E_b . The production function for A is shifted leftward by the initial transfer, to that shown by P_a'' , but the attainable equilibrium along this function at E_a'' is superior in utility terms to E_a , the anarchistic result. Upon this transfer, B will agree to respect the assigned own-product of A and A will agree to similarly respect the assigned own-product of B. Positive rights may be established, once the initial transfer has taken place to bring the two parties into a setting where the direct-production assignment is, in fact, Pareto-superior to anarchistic equilibrium.

Despite the extremely simple and abstract nature of the geometrical models presented, the conclusions are significant for an understanding of the conceptual emergence of individual rights. The analysis demonstrates that there is no necessary basis for any initial agreement that will simply acknowledge the rights of persons to retain those stocks of goods that they can wrest from the natural environment by their own labor. Something other than the utility function employed in standard economic theory must be introduced in order to provide an explanatory foundation for a structure of property law that legitimizes individuals' (families') claims to stocks actually produced by their own efforts and independently from interference from others. Nowhere in the analysis am I denying the possible existence of internal behavioral constraints that may serve to inhibit man's seizing stocks of goods produced by others or invading physical domain initially inhabited by others. I remain agnostic on this as on many other aspects of human nature. My emphasis here is that such constraints, if they do exist, are over and beyond those normally introduced in economic behavioral models. With this proviso, the result stated

becomes important. To secure an initial agreement on positive claims to goods or to resource endowments, some transfer of goods or endowments may be required. That is to say, some “redistribution” of goods or endowments may have to take place before a sufficiently acceptable base for property claims can be established. As the simple two-person model indicates, there may be many such redistributions that will meet the minimal requirements. Once any of these transfers takes place, if one is required, and/or behavioral limits are mutually accepted, positive rights of persons in stocks of goods or in resource endowments capable of producing goods may be settled. From this base, trades and exchanges in the postconstitutional stage already discussed can be implemented. These trades may, in utility terms, shift all persons to positions that overwhelmingly dominate either the natural equilibrium in anarchy or that distribution of goods and endowments that is settled on the initial establishment of positive individual rights.

Defection and Enforcement

To this point, attention has been concentrated on the conceptual bases for the formation of an initial social contract. The analysis has been aimed at isolating and identifying the mutuality of gain to be secured from a primal disarmament agreement accompanied, if necessary, by some unilateral transfers of goods or endowments. In this initial inclusive contract, all parties gain from the potential elimination of socially wasteful outlays on defense and predation. At the immediate postcontract stage, persons claim positive rights in stocks of goods, in resource endowments, and in specific spheres of activity. To this point, we have implicitly assumed that the set of rights agreed to will be respected by all participants.

This assumption cannot, of course, be justified. Even at this most elementary level of examination, the problem of enforcing contractual agreements must be introduced. Straightforward utility maximization will lead each person to defect on his contractual obligation if he expects to be able to accomplish this unilaterally. This may be illustrated in figure 4.3 (a duplicate of figure 2.1), which presents a two-by-two matrix for the two-person example. We are interested only in the net payoffs received by each of the two parties, A and B, in each of two possible positions. Each party has two behavioral options; he may keep his agreement, which amounts to respecting the de-

		B	
		Respects Rights 0	Respects No Rights V
A	Respects Rights 0	Cell I 19, 7	Cell II 3, 11
	Respects No Rights V	Cell III 22, 1	Cell IV 9, 2

Figure 4.3

financed rights of the other person. This is the action indicated by the 0 row and column of the matrix. Or, alternatively, each person may abide by no agreement and act strictly in narrow self-interest. This option is defined by the V row and column. If both persons take the V option, and refuse to abide by contracts made, the result is equivalent to that which was described earlier as the natural anarchistic equilibrium. If both persons respect the terms of contract, both are better off, and the 00 result in figure 4.3 represents the contractually agreed-on set of rights discussed earlier.

The numbers in the cells are utility indicators for the two persons, with the left-hand numbers indicating utility levels attainable for A, the right-hand numbers those attainable by B. As the numbers indicate, each person has an incentive to defect on the agreement provided that he expects to be able to do so unilaterally. If A defects, while B respects A's rights, the result is in Cell III, which is the most preferred of all positions shown for A. Similarly, if B defects, while A respects B's rights, a Cell II result emerges, which is the most favorable of all positions for B. The situation is analogous to the classical prisoners' dilemma in game theory.¹² Any positive structure of rights is, therefore, extremely vulnerable to defection if continued adherence to the contractual basis depends on voluntary and independent "law-abiding." In our illustration, A can obtain three units of utility by defaulting unilaterally from the result in Cell I; individual B can gain four units by defaulting uni-

12. For a generalized discussion which extends the dilemma to many social interactions, see Gordon Tullock, *The Social Dilemma*.

laterally and securing a result in Cell II. And if both persons defect, the system lapses back into a Cell IV outcome, and ultimately to the anarchistic equilibrium discussed.

In the simplified two-person interaction illustrated, however, it is surely plausible to suggest that rationality precepts will direct each person to adhere to the initial contractual terms. Each person will recognize that unilateral defection cannot succeed and that any attempt to accomplish this would plunge the system back into a position that is less desirable for everyone than that which is attained upon adherence to contract. As the payoffs or utility indicators in figure 4.3 suggest, neither A nor B would allow the other person to defect and get away with it. Once a defection has occurred, the other party can improve his own position by bringing the system back to the Cell IV position.

It is important to recognize explicitly the behavioral motivation that lends stability to the contractual solution in the two-person setting. Each person may respect the agreed-on assignment because he predicts that defection on his part will generate parallel behavior by the other party. Each person realizes that his own behavior influences the subsequent behavior of the other person and does so directly.

It is precisely this aspect of the interaction that is modified, in kind, as we shift from a two-person to a many-person setting. As more parties are added to the initial contractual agreement, in which an assignment of rights is settled, the influence of any one person's behavior on that of others becomes less and less. As an element inhibiting individual defections on an initial contract, this influence tends to disappear completely after some critical group size is reached. In large-number groups, each individual rationally acts as if his own behavior does not influence the behavior of others. He treats others' behavior as a part of his natural environment, and he adjusts his behavior accordingly. In this large-number setting, man ceases to be a "social animal" at least in this explicit behavioral sense. This setting remains analogous to an n -person prisoners' dilemma, but it is one in which fully voluntary compliance with contract, or law in any form, cannot be predicted. Each person has a rational incentive to default; hence, many persons can be predicted to default and the whole agreement becomes void unless the conditions of individual choice are somehow modified.¹³

13. The divergence between individual utility-maximization and group interest is char-

This relationship between voluntary adherence to mutually accepted rules of social interaction, whether these be ethical standards or property-rights assignments, and the size of the interacting group is familiar, but it does have specific relevance to our analysis here.¹⁴ The problem of enforcing any original contract becomes more difficult in large than in small groups. Any set of property rights, any legal structure, becomes more vulnerable to violation, and hence requires more than proportionate outlay on enforcement in large groups than in small. In respect to the conceptual origins of law and contract, this relationship alone suggests that contractual or quasi-contractual arrangements commence among individuals (families) that are involved in relatively small-number settings, with movement toward more inclusive contractual order taking the form of arrangements among smaller groups. These complexities are important, but they need not occupy our attention here.

If individual parties to an initial contract in which property assignments are established mutually acknowledge the presence of incentives for each participant to default and, hence, recognize the absence of viability in any scheme that requires dependence on voluntary compliance, they will, at the time of contract, enter into some sort of enforcement arrangement. Individuals' claims to stocks of goods and endowments will be accompanied by some enforcement institution that will be aimed to secure such claims.¹⁵ The nature of this enforcement contract or institution must be carefully examined. Each person will receive some benefit from the assurance that his established claims will be honored by others in the community. And there are mutual gains to all parties from engaging in some joint or collectivized enforcement effort. Enforcement of property claims, of individuals' rights to carry out designated activities, qualifies as a "public good" in the modern sense of this term.

Enforcement is, however, different from the more familiar examples of

acteristic of many other situations in addition to that which involves adherence to law in the narrow sense of the term.

14. For a general discussion, see my "Ethical Rules, Expected Values, and Large Numbers," *Ethics* 76 (October 1965): 1–13.

15. The necessity for including enforcement provisions in the initial agreement distinguishes the social contract from other contracts which are made within the framework of a legal order. John C. Hall emphasizes that the recognition of this feature of social contract is central to the ideas of both Hobbes and Rousseau. See John C. Hall, *Rousseau: An Introduction to His Political Philosophy* (London: Macmillan, 1973), pp. 86–92.

public goods in several essential respects.¹⁶ In order to be effective, enforcement must include the imposition of physical constraints on those who violate or attempt to violate the rights structure, on those who break the law. It is this characteristic that creates problems. There is no obvious and effective means through which the enforcing institution or agent can itself be constrained in its own behavior. Hence, as Hobbes so perceptively noted more than three centuries ago, individuals who contract for the services of enforcing institutions necessarily surrender their own independence.

Consider, say, a hundred-man community. In the absence of enforcement, let us say that B violates the contract that settles all property claims. He does so by stealing goods from or by interfering with the designated personal liberties of A. The latter will, of course, have some incentive to react privately by a counterattack on B. But if this becomes the general pattern of behavior, the system rapidly degenerates toward the precontractual position of anarchistic equilibrium.¹⁷ Individuals C, D, E, . . . , however, have no direct interest in punishing B for stealing from or interfering with A. They have an indirect interest insofar as such punishment makes their own claims more secure, but unless they make such a connection in their own conception of enforcement, they may be reluctant to approve particularized punishment. This problem may be handled by an agreement by all persons on the purchase of the services of some external enforcing agent or institution that will, in all particular cases, take the enforcing-punishment action required. The "public good" is the generalized security of rights or claims, and not the particular enforcement action which produces this security.

In an idealized sense, the enforcing institution is necessarily external to the parties that reach agreement in the initial contract. The analogy to a simple game may be helpful. Two boys mutually acknowledge some division of marbles between them, and they seek to play a game. Each boy may know, however, that his opponent will have a strong incentive to cheat unless he is closely monitored. They agree and appoint a referee or umpire, inform him about the specific rules under which they choose to play, and ask that he en-

16. This point will become clear in the more general discussion of Chapters 7 and 8.

17. Historically, the state's more general role in criminal law emerged from its early role as enforcer of the process within which private quarrels were settled, by physical combat or otherwise. See Henry Maine, *Ancient Law*.

force adherence to these designated rules. This is precisely the functional role assigned to the state in its law-enforcement task. The state becomes the institutionalized embodiment of the referee or umpire, and its only role is that of insuring that contractual terms are honored.

This analogy exposes a recurrent fallacy in many discussions of property rights and of the role of the state in enforcing these rights. Enforcement of claims is categorically different from defining these claims in the first place. Claims are conceptually agreed upon by all parties in the constitutional stage of social contract. The state is then called upon to monitor these claims, to serve as an enforcing institution, to insure that contractual commitments are honored. To say that rights are defined by the state is equivalent to saying that the referee and not the players chooses both the initial division of the marbles and the rules of the game itself.

The Protective State and the Productive State

The distinction between the constitutional and the postconstitutional stages of social contract allows us to interpret the state, the collective agency of the community, in two separate roles. Failure to keep these roles distinct, in theory or in practice, has produced and continues to produce major confusion. At the constitutional stage, the state emerges as the enforcing agency or institution, conceptually external to the contracting parties and charged with the single responsibility of enforcing agreed-on rights and claims along with contracts which involve voluntarily negotiated exchanges of such claims. In this “protective” role, the state is not involved in producing “good” or “justice,” as such, other than that which is embodied indirectly through a regime of contract enforcement. Explicitly, this state cannot be conceived as some community embodiment of abstract ideals, which take form over and beyond the attainment of individuals. This latter conception is and must be foreign to any contractarian or individualistic vision or model of social order. Nonetheless, because of each person’s interest in the security of his agreed-on rights, the legal or protective state must be characterized by precepts of neutrality. Players would not consciously accept the appointment of a referee who was known to be unfair in his enforcement of the rules of the game, or at least they could not agree on the same referee in such cases.

“Fairness” or “justice” may emerge, therefore, in a limited sense from the self-interest of persons who enter the enforcement contract. It will not emerge from the acceptance of overriding ideals for society at large.

This legal or protective state, the institutions of “law” broadly interpreted, is *not* a decision-making body. It has no legislating function, and it is not properly represented by legislative institutions. This state does not incorporate the process through which persons in the community choose collectively rather than privately or independently. The latter characterizes the functioning of the conceptually separate productive state, that agency through which individuals provide themselves with “public goods” in postconstitutional contract. In this latter context, collective action is best viewed as a complex exchange process with participation among all members of the community. This process is appropriately represented by legislative bodies and the decision-making, choosing process is appropriately called “legislation.” By sharp contrast, the protective state which carries out the enforcement task assigned to it in constitutional contract makes no “choices” in the strict meaning of this term. Ideally or conceptually, enforcement might be mechanically programmed in advance of law violation. The participants agree on a structure of individual rights or claims that is to be enforced, and violation requires only the findings of fact and the automatic administration of sanctions. A contract or a right is or is not violated; this is the determination to be made by “the law.” Such a determination is not “choice” in the classic sense that the benefits of one alternative are weighed against opportunity costs (the benefits foregone). “The law,” enforced by the state, is not necessarily that set of results which best represents some balance of opposing interests, some compromise, some median judgment. Properly interpreted, “the law” which is enforced is that which is specified to be enforced in the initial contract, whatever this might be.

I am not, of course, suggesting that ambiguities are wholly absent or that the actual enforcement task of the state is purely mechanistic. These aspects should not, however, distract attention from the characteristic and central feature of the enforcement contract, which is designed to implement the detection of violation and the punishment of violators of explicitly accepted and well-defined rights and claims. As noted, “the law” steps beyond the bounds of propriety when it seeks, and explicitly, to redefine individual rights.

If, indeed, the state is conceived in this sense, genuine choice is involved since the benefits and costs of various schemes for redefinition would have to become relevant.

From nothing more than this brief and introductory discussion, much of the modern confusion can be appreciated. Appropriately, the judiciary, as an element of the enforcement structure, is independent of the choice-making arm of the collectivity, the legislature. However, as the judiciary itself violates the terms of its own contract by explicitly engaging in legislation, in genuine “social choice,” its independence from choice-making rules has been properly brought into question. The legal or protective state, as such, is not “democratic” in the sense that collective decisions are reached through some voting process, whether this be majority voting or otherwise. In determining the facts of contractual agreement, plural rather than individual or unitary judgments may be invoked, and these may be combined in many ways. In many jurisdictions, all members of a jury must be in agreement before a verdict can be established. Appellate courts may require only a simple majority. In all such instances, however, it should be clear that plurality rules are nothing other than devices aimed at producing somewhat more accuracy in a final finding of fact. With genuine choice-making, “accuracy” is not an appropriate descriptive word. Genuine collective choice may be rational or irrational; benefits and costs may or may not be properly weighed one against the other. But choices cannot, in themselves, be accurate or inaccurate, since it is values that are at issue, not facts.

Rules as Indirect Rights

To this point in the discussion of constitutional contract, we have assumed that agreement is reached on the limits to behavioral interaction and on the positive set of claims to endowments of goods, accompanied by some enforcement contract with the protective state. In an all-private-goods world, this would be the end of it. Trades and exchanges among persons in postconstitutional stages would more or less naturally emerge, as discussed in Chapter 3. However, when we allow for the presence of jointly shared collective or public goods and services, the collectivity as the productive state and its rules for operation must be taken into account. The political constitution, which in our context is only one aspect of the broader constitutional contract, be-

comes important here, and the rules for making collective decisions concerning the provision and the cost-sharing of public goods must, themselves, be settled at the ultimate constitutional stage of negotiation. It would, as noted in Chapter 3, be of relatively little moment to define an individual's nominal claims to goods only to leave these claims fully vulnerable to unconstrained political exploitation.

An earlier work, *The Calculus of Consent*, written jointly with Gordon Tullock, was devoted largely to an analysis of the constitutional choice among rules for making collective decisions. In that analysis, Tullock and I assumed implicitly that individual participants in constitutional deliberations over alternative rules faced uncertainty concerning their own interests in future collective decisions. Nonetheless, we did not question the independent establishment of their ultimate rights and claims to property, human and non-human, beyond the range of collective-decision rules. As I have suggested, this approach was an extension and application of orthodox economic methodology, which has tended to neglect the critical problems of establishing individual rights. This book differs from *The Calculus of Consent* in this fundamental respect; here I am trying to analyze the initial contract that assigns rights and claims among persons. This difference allows collective-decision rules to be interpreted in a somewhat modified setting, namely, as an integral part of a more inclusive contract rather than a strictly political constitution superimposed on some previously negotiated settlement. In the earlier book, we argued that the criterion of acceptability or efficiency lay in agreement, in unanimity. Further, we argued that insofar as participants remain uncertain as to their own specific roles in subsequent operation under the rules chosen, they would tend to reach agreement on reasonably "fair" and "efficient" working rules.¹⁸ We did not postulate initial equality among individuals in property rights or in capacities, but our presumption of uncertainty served to generate a plausible basis for agreement on rules for collective action.

In the model incorporated here, by comparison, I allow quite explicitly

18. In this respect, as noted earlier, our approach had considerable affinity with that of John Rawls, who has attempted to derive general principles of justice in a similar manner. Rawls's earlier papers appeared in the 1950s, but his work is presented in detail in his book *A Theory of Justice*.

for personal inequality in the natural equilibrium, the anarchistic base from which primal disarmament contracts are conceptually negotiated. As the analysis in the earlier parts of this chapter indicated, however, the establishment of positive claims to stocks of goods or endowments may not be possible until and unless some unilateral transfers are made. This potential for transfer allows us to introduce an additional dimension of adjustment which may possibly facilitate the reaching of agreement among parties in contract. When we recognize that the rules for collective decision-making at the postconstitutional stage are also to be settled as a part of the initial contract, we have available yet another dimension for adjustment.

Consider the calculus of an individual whose position in anarchistic equilibrium is not significantly worse than that which he expects to secure under a simple disarmament agreement. When he also recognizes the problems of enforcement, including those which involve constraining the enforcing agent, this individual may be quite reluctant to enter into the basic social contract at all. Suppose, however, that one of the many clauses in a proffered contractual settlement states that “public goods” are to be financed by progressive income taxes, and that the person in question has either higher-than-average expected demands for public goods or lower-than-average income-wealth expectations. This proposed part of the larger social contract now represents, for this person, a positive supplement to the set of claims that he might otherwise secure from the unamended disarmament agreement. The collective decision rules present him with something akin to additional “rights” and upon which he may place a positive value. He may be motivated to enter into the constitutional contract under such conditions, even without a unilateral transfer of goods, although this would be an alternative means of making the proposal attractive to him.

The Constitutional Mix

The inclusive constitutional contract embodies elements that may appear in alternative combinations or mixes. The terms must include, first of all, some statement of limits on the behavior of any person with respect to the positions of other persons in the community. This element was referred to earlier in this chapter as the disarmament contract. As they enter genuine society from anarchy, persons lay down their arms; they accept rules governing their

own behavior in exchange for the like acceptance of such rules on the part of others. Secondly, the basic contract must define the positive rights of possession or domain over stocks of goods, or more generally, over resource endowments capable of producing final goods. These endowments include human capacities (the rights to one's own person which have been widely discussed in the theory of property), as well as nonhuman factors, including domain over territory. These ownership rights or claims may simply reflect the pattern of possession established directly when interpersonal interferences are eliminated, which we have called the direct-production imputation, but, as the analysis disclosed, certain "exchanges" of resource endowments or goods and behavioral constraints may be necessary before clearly acknowledged ownership imputations are possible. Along with the limits on behavior and the rights of ownership, the inclusive constitutional contract must also make explicit the terms and conditions of enforcement. This set of terms will specify in detail the operation and limits of the protective state that is established as the enforcing agent. Finally, the basic contract must define the rules under which the collectivity must operate in making and in implementing decisions concerning the provision and financing of "public goods." This set of terms will specify in detail the operation and the limits of the productive state, the legislative aspect of collective organization. The rules and institutions of this productive state may, in themselves, incorporate several dimensions. The contract should indicate the allowable range over which collective action may take place. That is to say, some restrictions on the type of goods to be provided and financed collectively must be included. At least in some rough sense, the dividing line between the private and the public or governmental sector of the economy should be settled in the basic constitution. Within these defined limits, allowable departures from unanimity in reaching collective decision should be specified. Such departures need not, of course, be uniform over all decisions. Institutions for cost-sharing, that is, tax institutions, may also be imbedded in the inclusive constitutional structure.

It is not my purpose here to develop criteria for efficiency in constitutional contract in any specific setting. The mix among the various elements in this inclusive settlement will be functionally related to several identifiable characteristics of the community of individuals. This will include the size of the membership itself as well as the environmental setting. The features of

the anarchistic natural equilibrium, whether or not this is ever actually realized, will influence the relative positions of individuals and groups in the final constitutional settlement. The degree as well as the distribution of the inequalities among persons will be important in this respect. Individuals may differ, and may be thought to differ, in relative capacities to produce goods and to secure gains by predation on their fellows. These differences, along with differences in individuals' tastes for productive and predatory activities, will have predictable effects on the initial settlement. Expectations about demands for publicly provided goods and services along with expectations about relative income and wealth levels will also affect individuals' willingness to accept rules for collective action.

The most significant point that emerges from this very general discussion is the interdependence among the several elements in the constitutional mix. Contrary to orthodox economic methodology, the rights of persons to property, the rights to do things privately and individually with physical resources, cannot be treated in isolation from those rights which are indirectly represented by membership in a collectivity that is constitutionally empowered to make decisions under predetermined rules. Consider, for example, the position of a person who holds nominal ownership rights to an income stream from a scarce and highly valued resource (human or nonhuman). This private ownership claim may be tempered by the membership rights in the collectivity, the governmental institutions of the community, that are held by other persons, membership rights that may offer other persons some indirect claims on the differentially higher income stream in question. This is not to suggest that the specific constitutional mix chosen need be the most efficient. As noted in Chapter 3, all parties might have gained by an initial transfer of claims with substantially greater stability of nominal ownership claims.

This approach allows us to look somewhat differently and in a positive manner at the perplexing issue of income-wealth redistribution. Under certain constitutional structures, those persons who are relatively "poor" do not properly claim, on the basis of overriding ethical norms, a share in the economic returns or assets of those who are relatively "rich." They *may* claim some such share indirectly on the basis of commonly held membership in collectively organized community under specified constitutional contract. The relatively "rich," in their turn, may legitimately expect their "private rights" to be respected and honored, and violations of these rights enforced,

only as a component part of the more inclusive contractual arrangement which predictably requires that they pay differentially higher shares in those goods and services provided jointly for the whole community.¹⁹ In this larger and more inclusive contract, all individuals and groups should find it advantageous to adhere to the rules established, to respect the claims as tempered, and to conduct themselves in such fashion as to attain maximum individual liberty within the constraints of acceptable order.

19. For a more detailed examination, see my paper "The Political Economy of the Welfare State," Research Paper No. 808231-1-8 (Center for Study of Public Choice, Virginia Polytechnic Institute and State University, June 1972).

See also Earl Thompson, "The Taxation of Wealth and the Wealthy" (UCLA Department of Economics Working Paper, February 1972).